NEET AIR (2027)

[Special Test Series]

TEST-02

DURATION: 180 MIN. | **DATE : 20/08/2026** | **M. MARKS: 720**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **720 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **45** questions in each subject
3. The test booklet consists of **180** questions. The maximum marks are **720**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics, 1-45)**, Section-II **(Chemistry, 46-90)**, Section-III **(Botany, 91-135)**, Section-IV **(Zoology, 136-180)** and having **45** questions in each part in which all **45** questions are of Objective Type and all **45** questions are **compulsory.**
5. There is only **one correct** response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
8. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. Use of white fluid for correction is **not permissible** on the **Answer Sheet**.
9. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the **OMR Sheet** is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

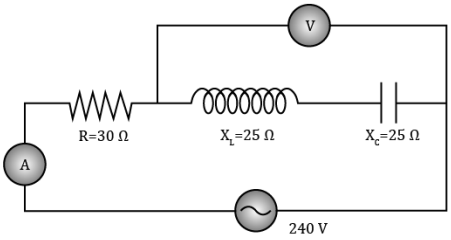
Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

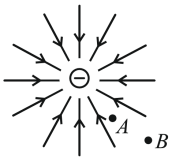
Candidate's Signature : _____ **Invigilator's Signature** _____

Q1 In the circuit shown in figure, neglecting source resistance, the voltmeter and ammeter reading will respectively be;



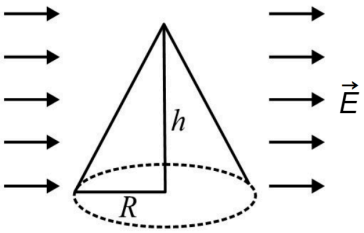
(1) 0V, 3 A
(2) 150 V, 3 A
(3) 150 V, 6 A
(4) 0 V, 8 A

Q2 Figure shows the field lines of a point negative charge. In going from B to A, the kinetic energy of a small negative charge will:



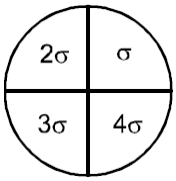
(1) increase
(2) decrease
(3) remain constant
(4) data insufficient.

Q3 In figure, a cone lies in a uniform electric field $\vec{E}$. Determine the magnitude of electric flux entering the cone.



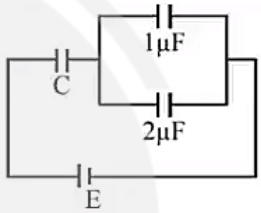
(1) $E\pi R^2$
(2) ERh
(3) $E\pi h^2$
(4) ER^2

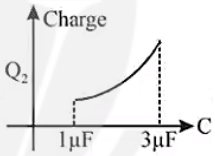
Q4 A disc of radius $\sqrt{3}$ R having uniformly distributed charges of charge densities σ, 2σ, 3σ and 4σ in first, second, third and fourth quadrant respectively. Electric potential at a distance R (from centre) on the axis of disc is:

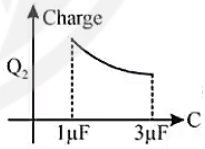


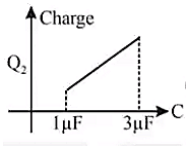
(1) $\frac{\sigma}{2\epsilon_0}R$
(2) $\frac{10\sigma}{4\epsilon_0}R$
(3) $\frac{\sigma R}{4\epsilon_0}$
(4) $\frac{5\sigma R}{4\epsilon_0}$

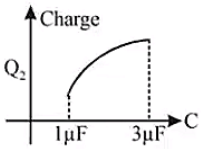
Q5 In the given circuit, charge Q_2 on the 2μF capacitor changes as C is varied from 1μF to 3μF. Q_2 as a function of C is given properly by : (figures are drawn schematically and are not to scale.)



(1) 

(2) 

(3) 

(4) 

Q6 Two wires each of radius of cross section r but of different materials are connected together end to end (in series). If the number densities of charge carriers in the two wires are in the ratio 1:4, the drift velocity of electrons in the two wires will be in the ratio.

(1) 1 : 2 (2) 2 : 1
(3) 4 : 1 (4) 1 : 4

Q7 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: If two particles each having equal mass and equal charge are projected with different speeds in region of same transverse magnetic field then their time period of revolution will be equal.

Reason R: If a charged particle is revolving in region of uniform transverse magnetic field then its time period is equal to $\frac{2\pi m}{qB}$ (where symbols have their usual meaning.)

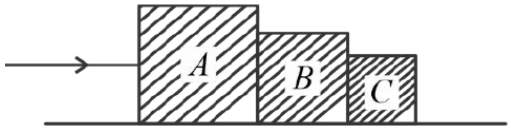
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A .
(4) Both A and R are true but R is NOT the correct explanation of A .

Q8 A particle moves along a straight line *OX*. At a time *t* (in second) the position *x* (in metre) of the particle from *O* is given by $x = 40 + 12t - t^3$. How long would the particle travel before coming to rest?

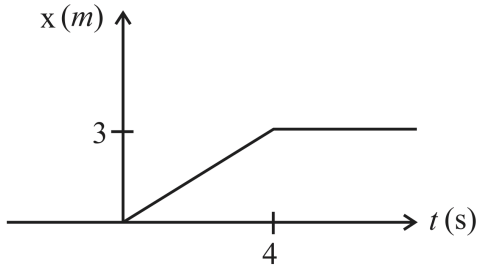
(1) 24 m (2) 40 m
(3) 56 m (4) 16 m

Q9 Three blocks *A*, *B* and *C* of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a frictionless surface, as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between *A* and *B* is



(1) 8 N (2) 18 N
(3) 2 N (4) 6 N

Q10 Figure shows the position-time graph of a particle of mass 4 kg. Let the force on the particle for $t < 0$, $0 < t < 4$s, $t > 4$ s be F_1, F_2 and F_3 respectively. Then



(1) $F_1 = F_2 = F_3 = 0$
(2) $F_1 > F_2 = F_3$
(3) $F_1 > F_2 > F_3$
(4) $F_1 < F_2 < F_3$

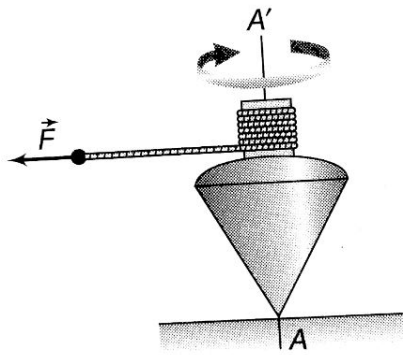
Q11 Write the dimensions of $\frac{a}{b}$ in the relation $P = \frac{a - t^2}{bx}$;where *P* is the pressure, x is the distance, and *t* is the time.

(1) $\left[M^{-1}L^0T^{-2}\right]$ (2) $\left[ML^0T^{-2}\right]$
(3) $\left[ML^0T^2\right]$ (4) $\left[MLT^{-2}\right]$

Q12 A moving block having mass *m*, collides with another stationary block having mass 4*m*. The lighter block comes to rest after collision. When the initial velocity of the lighter block is *v*, then the value of coefficient of restitution (*e*) will be:

(1) 0.8 (2) 0.4
(3) 0.5 (4) 0.25

Q13 The top in figure has a moment of inertia of 4.00×10^{-4} kg. m^2 and is initially at rest. It is free to rotate about the stationary axis *AA'*. A string, wrapped around a peg along the axis of the top, is pulled in such a manner as to maintain a constant tension of 2.5 N. If the string does not slip while it is unwound from the peg, what is the angular speed of the top after 80.0 cm of string has been pulled off the peg?



(1) 75 rad/s (2) 50 rad/s
(3) 125 rad/s (4) 100 rad/s

Q14 Given below are three statements. Based on their truthness/falseness, choose **correct** option.

Statement I: Angular momentum of planets revolving around sun is conserved.

Statement II: The total energy of a satellite orbiting the earth is negative.

Statement III: Linear momentum of planets revolving around sun is also conserved.

(1) Statement I, Statement II are incorrect, Statement III is correct.
(2) Statement I, Statement II are correct, Statement III is incorrect.
(3) Statement I, Statement III are incorrect, Statement II is correct.
(4) Statement II, Statement III are correct, Statement I is incorrect.

Q15 An asteroid of mass m is approaching earth, initially at a distance $10R_E$ with speed v_i. It hits earth with a speed v_f (R_E and M_E are radius and mass of earth), then

(1) $v_f^2 = v_i^2 + \frac{2GM_E}{R_E}\left(1 + \frac{1}{10}\right)$
(2) $v_f^2 = v_i^2 + \frac{GM_E}{R_E}\left(1 + \frac{1}{10}\right)$
(3) $v_f^2 = v_i^2 + \frac{2GM_E}{R_E}\left(1 - \frac{1}{10}\right)$
(4) $v_f^2 = v_i^2 + \frac{GM_E}{R_E}\left(1 - \frac{1}{10}\right)$

Q16 Two wires of different material have same length and area of cross section. What is the ratio of their increase in length $\left(\frac{\Delta L_1}{\Delta L_2}\right)$, when the forces applied are the same?

$\left(Y_1 = 0.90 \times 10^{11}\ \mathrm{Nm}^{-2}\right)$
$\left(Y_2 = 3.60 \times 10^{11}\ \mathrm{Nm}^{-2}\right)$

(1) 4 : 1 (2) 1 : 4
(3) 2 : 1 (4) 1 : 2

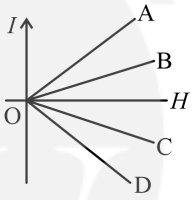
Q17 An open organ pipe of length *l* is sounded together with another organ pipe of length *l* + *x* in their fundamental tones (*x* << *l*). The beat frequency heard will be (speed of sound is *v*):

(1) $\frac{vx}{4l^2}$ (2) $\frac{vl^2}{2x}$
(3) $\frac{vx}{2l^2}$ (4) $\frac{vx^2}{2l}$

Q18 The work done in turning a magnet of magnetic moment '*M*' by an angle of 90° from the meridian is 'n' times the corresponding work done to turn it through an angle of 60°, where 'n' is given by:

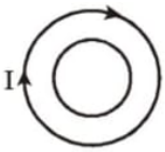
(1) 1/2 (2) 2
(3) 1/4 (4) 1

Q19 The variation of the intensity of magnetisation (*I*) with respect to the magnetising field (*H*) in a diamagnetic substance is described by the graph.



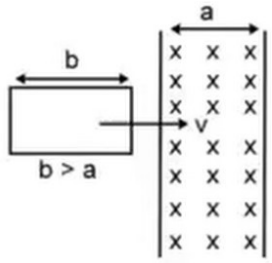
(1) OC (2) OD
(3) OA (4) OB

Q20 Two different wire loops are concentric and lie in the same plane. The current in the outer loop (I) is clockwise and increases with time. The induced current in the inner loop:



(1) is clockwise.
(2) is zero.
(3) is counter clockwise.
(4) has a direction that depends on ratio of the loop radii.

Q21 In the given arrangement, the loop is moved with constant velocity v in a uniform magnetic field B in a restricted region of width a. The time for which the emf is induced in the circuit is:



(1) $\frac{2b}{v}$
(2) $\frac{2a}{v}$
(3) $\frac{(a+b)}{v}$
(4) $\frac{2(a-b)}{v}$

Q22 An object of length 6 cm is placed perpendicularly on the principal axis of a concave mirror of focal length *f* at a distance of 4*f*. The length of the image will be:
(1) –2 cm
(2) 12 cm
(3) 4 cm
(4) 1.2 cm

Q23 An electromagnetic radiation has an energy of 13.2 keV. Then, the radiation belongs to the region of:
(1) Visible light. (2) Ultraviolet.
(3) Infrared. (4) X-ray

Q24 Unpolarised light falls on two polarising sheets placed one on top of the other. What must be the angle between the characteristic directions of the sheets, if the intensity of the final transmitted light is one third the maximum intensity of the first transmitted beam?
(1) $\cos^{-1}\left(\frac{1}{4}\right)$ (2) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
(3) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (4) $\cos^{-1}\left(\frac{1}{3}\right)$

Q25 A fish is near the centre of a spherical fish bowl filled with water $\left(\mu = \frac{4}{3}\right)$. A child stands in air at a distance 2*R* (*R* is the radius of curvature of the sphere) from the centre of the bowl. At what distance from the centre would the child's nose appear to the fish situated at the centre?
(1) R (2) 2R
(3) 3R (4) 4R

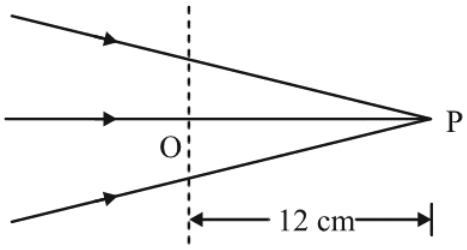
Q26 Work-function for cesium metal is 2.14 eV. Let a beam of light of frequency 6×10^{14} Hz is incident over the metal surface.
Now, match the following columns and choose the correct option from codes given.

List-I		List-II	
(A)	Maximum KE of emitted photoelectrons (in eV)	(I)	332.3 kms^{-1}
(B)	Minimum KE of emitted photoelectrons (in eV)	(II)	345 mV
(C)	Stopping potential of material (in mV) is	(III)	0.345 eV
(D)	Maximum speed of the emitted photoelectrons (in kms^{-1})	(IV)	0

(1) A-IV, B-III, C-II, D-I
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

Q27 In Young's double-slit experiment, the slits are 2mm apart and are illuminated by photons of two wavelengths λ_1= 12000Å and λ_2=10000Å. Minimum distance from the common central bright fringe on the screen 2 m from the slit will a bright fringe from one interference pattern coincide with a bright fringe from the other?
(1) 4 mm
(2) 3 m
(3) 8 mm
(4) 6 mm

Q28 Figure given below shows a beam of light converging at point *P*. When a concave lens of focal length 16 cm is introduced in the path of the beam at a place *O* shown by dotted line such that *OP* becomes the axis of the lens, the beam converges at a distance *x* from the lens. The value of *x* will be equal to:



(1) 12 cm
(2) 24 cm
(3) 36 cm
(4) 48 cm

Q29 Ratio of wavelengths emitted for transition 2 $\rightarrow$1 in Li^{2+}, He^{+} and H is:
(1) 1 : 2 :3 (2) 1 : 4 : 9
(3) 4 : 9 : 36 (4) 3 : 2 :1

Q30 The binding energy per nucleon of deuterium and helium nuclei are 1.1 MeV and 7 MeV respectively. When two deuterium nuclei fuse to form a helium nucleus the energy released in the fusion is:
(1) 23.6 MeV (2) 25.1 MeV
(3) 22.3 MeV (4) 25.8 MeV

Q31 From a height of h above the ground, a ball is projected up at an angle 30° with the horizontal. If the ball strikes the ground with a speed of 1.25 times its initial speed of 40 m s^{-1}, the value of *h* is (acceleration due to gravity = 10 m s^{-2}).
(1) 75 m (2) 60 m
(3) 30 m (4) 45 m

Q32 A ball is attached to end of string of length which is initially vertical shown.
List-I shows the values of speeds related to the ball.
List-II shows the required motion in the vertical circle. Match **List-I** with **List-II**:

List-I		List-II	
(A)	$u \geq \sqrt{5gl}$	(I)	Oscillation of the ball
(B)	$u < \sqrt{2gl}$	(II)	Complete the vertical circle
(C)	$\sqrt{2gl} < u < \sqrt{5gl}$	(III)	Speed at highest point when thrown with critical speed at lowest point
(D)	$\sqrt{gl}$	(IV)	Ball crosses the horizontal position but does not complete the circle

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-I, C-II, D-IV

Q33 A hemispherical bowl just floats without sinking in a liquid of density 1.2×10^3 kg/m^3. If outer diameter and the density of the bowl are 1 m and 2×10^4 kg/m^3 respectively then the inner diameter of the bowl will be:
(1) 0.94 m (2) 0.97 m
(3) 0.98 m (4) 0.99 m

Q34 The temperature of 5 moles of a gas which was held at constant volume was changed from $100^{\circ}C$ to $120^{\circ}C$. The change in internal energy was found to be 80 joules. The total heat capacity of the gas at constant volume will be equal to:
(1) $8J/K$
(2) $0.8\ J/K$
(3) $4.0\ J/K$
(4) $0.4\ J/K$

Q35 One solid sphere, having mass 1 kg and diameter 0.3 m is suspended from a wire. If the twisting couple per unit twist for the wire is $6 \times 10^{-3} N-m/$ radian, then the time period of small oscillations will be
(1) 6.28 s (2) 7.7 s
(3) 8.28 s (4) None of these

Q36 A particle of mass *m* is executing oscillations about the origin on the *x*-axis with amplitude *A*.

Its potential energy is given as $U(x) = Bx^4$, where B is a positive constant. The x-coordinate of the particle, where the potential energy is one-third of the kinetic energy, is:

(1) $\pm\frac{A}{2}$ (2) $\pm\frac{A}{\sqrt{2}}$

(3) $\pm\frac{A}{3}$ (4) $\pm\frac{A}{\sqrt{3}}$

Q37 Two metallic spheres P and Q are of same surface finish and same material. Weight of P is twice that of Q. Both spheres are heated to same temperature and are left in room to cool by radiation. The ratio of initial rate of cooling of P to that of Q is:

(1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$

(3) $1:2^{\frac{1}{3}}$ (4) $2^{\frac{2}{3}}:1$

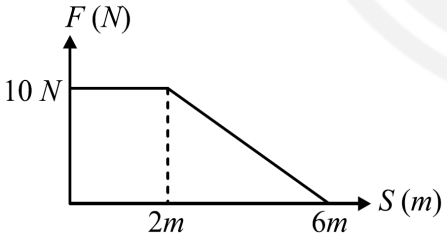
Q38 The rms velocity of a gas kept at a temperature of 300 K in a vessel is found to be 500 m/s. The vessel is filled with which gas?

(Take R = 8 J/mol-K for easy calculation)

(1) O_2 (2) N_2

(3) H_2 (4) CO_2

Q39 A body of constant mass $m = 1$ kg moves under variable force F as shown. If at $t = 0$, $S = 0$ and velocity of the body is $\sqrt{20}$ m/s and the force is always along direction of velocity, then choose the correct option



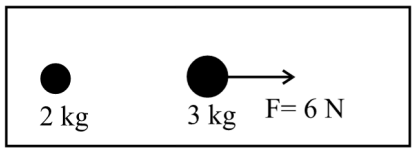
(1) Velocity of the particle will increase upto $S = 2m$ and then decrease.

(2) The final velocity at $S = 6$ m is 10 m/s

(3) The final velocity at $S = 6$ m is $4\sqrt{5}$ m/s

(4) The acceleration is constant up to $S = 2$m and then it is negative.

Q40 Two particles are shown in the figure. At $t = 0$ a constant force $F = 6$N starts acting on the 3kg particle. Find the velocity of the centre of mass of these particles at $t = 5$ s.



(1) 5 m/s (2) 4 m/s

(3) 6 m/s (4) 3 m/s

Q41 Match **List-I** (devices) with **List-II** (their applications)

List-I		List-II	
(A)	A.C. generator	(I)	Detect the presence of small current in circuit.
(B)	Light emitting diode	(II)	Change input A.C voltage to higher or smaller value
(C)	Transformer	(III)	Convert mechanical energy into electrical energy
(D)	Galvanometer	(IV)	Convert electrical energy into light energy

(1) A-IV, B-III, C-II, D-I

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-II, C-III, D-I

(4) A-III, B-IV, C-II, D-I

Q42 A particle is moving three times as fast as an electron. The ratio of de-Broglie wavelength of the particle to that of electron is 1.81×10^{-4}. Calculate the particle's mass.

(1) 1.275×10^{-26} kg

(2) 1.576×10^{-27} kg

(3) 1.765×10^{-25} kg

(4) 1.675×10^{-27} kg

Q43 A rectangular wire loop of side lengths a and b carries a steady current I. The magnetic field intensity at the centre of the rectangle (i.e. the point of intersection of its diagonals) is:

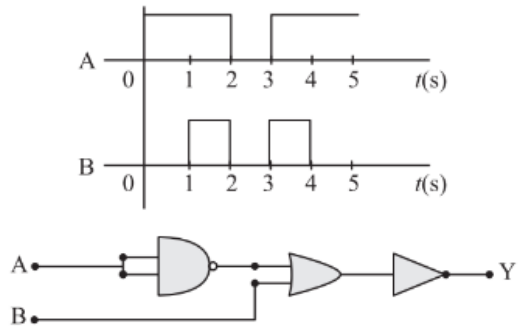
(1) $\frac{2\mu_0 I}{\pi}\left(\frac{\sqrt{a^2+b^2}}{ab}\right)$

(2) $\frac{4\mu_0 I}{\pi}\left(\frac{\sqrt{a^2+b^2}}{ab}\right)$

(3) $\frac{4\sqrt{2}\mu_0 I}{\pi}\left(\frac{\sqrt{ab}}{a^2+b^2}\right)$

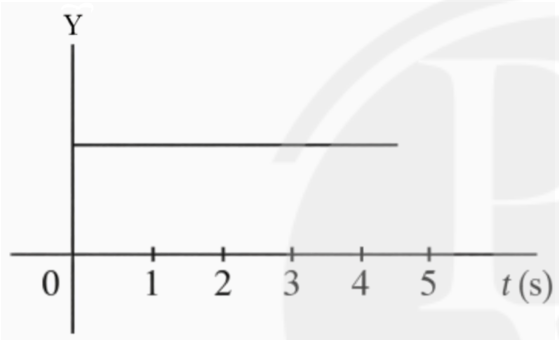
(4) $\frac{8\sqrt{2}\mu_0 I}{\pi}\left(\frac{\sqrt{ab}}{a^2+b^2}\right)$

Q44 Draw the output signal Y in the given combination of gates.

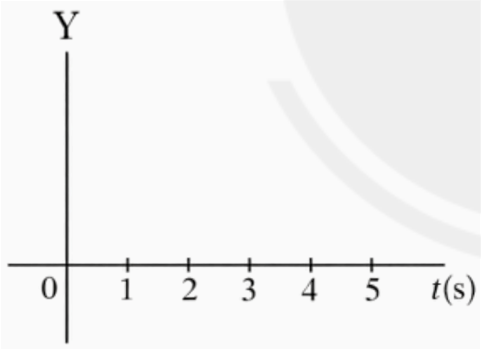


[JEE Main 2021, 26 Feb (Shift 2)]

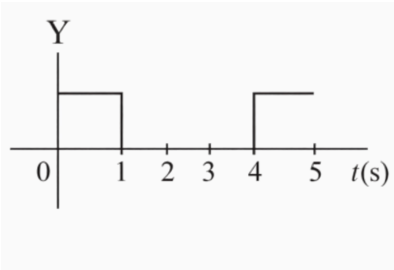
(1)



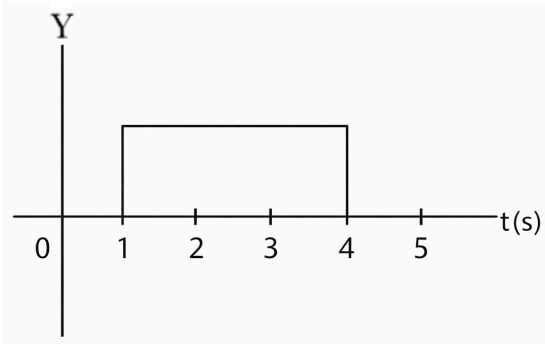
(2)



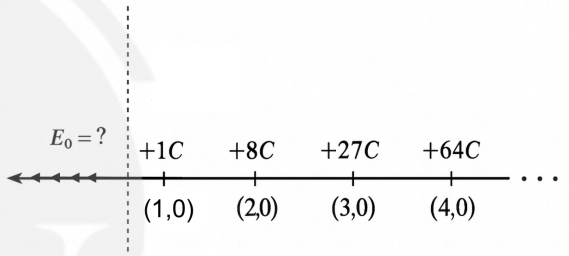
(3)



(4)



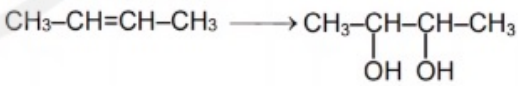
Q45 Ten point charges are placed on the x-axis at positions (1,0),(2,0),(3,0),...,(10,0), such that at position $(n, 0)$ the charge is n^3 coulombs (i.e., 1^3C, 2^3C, ..., $10^3$1C).If the net electric field magnitude at the origin $(0, 0)$ due to the ten charges is 'NK' V/m ,K is Coulomb constant ,then value of N is :



(1) 45 (2) 35

(3) 55 (4) 50

Q46 For the following reaction:



Reagent is :

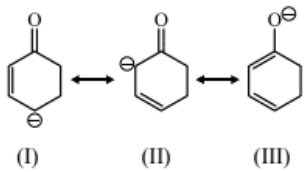
(1) HCl

(2) Dil. H_2SO_4

(3) Bayers reagent

(4) H_2/Ni

Q47 Decreasing order of stability among the Resonance Structures is:

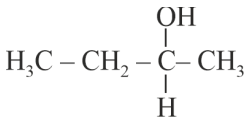


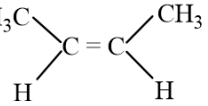
(1) (I) > (II) > (III)

(2) (III) > (I) > (II)
(3) (II) > (I) > (III)
(4) (III) > (II) > (I)

Q48 What is the major product (A) of the following reaction?

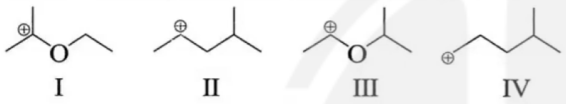
$$CH_3 - C \equiv C - CH_3 + H_2 \xrightarrow[(2)\ BH_3, H_2O_2]{(1)\ Na\ /\ liquid\ NH_3} A$$

(1) 

(2) 

(3) $CH_3CH_2CH_2CH_3$
(4) $CH_3 - CH_2 - CH_2 - CH_2 - OH$

Q49 The correct stability order for the following species is:



(1) II > IV > I > III
(2) I > II > III > IV
(3) II > I > IV > III
(4) I > III > II > IV

Q50 Consider the following reaction at $600^{\circ}C$.

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g);$$

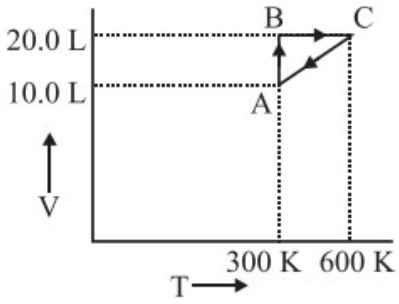
$K_C = 4.00$

A mixture was prepared with $[SO_3] = 0.50M$, $[O_2] = 0.35M$ and $[SO_2] = 0.0M$

Thus, equilibrium concentration of O_2 is

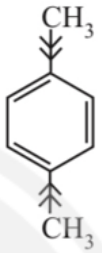
(1) 0.0450 M (2) 0.035 M
(3) 0.35 M (4) 0.45 M

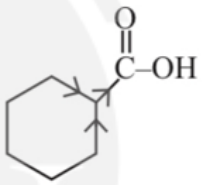
Q51 This graph expresses the various steps of the system containing 1 mole of gas. Which type of process system has when it moves from C to A ?

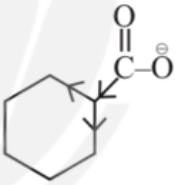


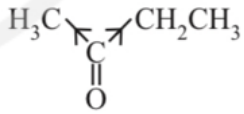
(1) Isothermal (2) Adiabatic
(3) Isobaric (4) Isochoric

Q52 In which of the following species, **incorrect** direction of inductive effect is shown?

(1) 

(2) 

(3) 

(4) 

Q53 The solubility of AgCl in 0.1 M NaCl is (K_{sp} of AgCl = 1.2×10^{-10})

(1) 0.05M
(2) 1.2×10^{-6}M
(3) 2×10^{-5}M
(4) 1.2×10^{-9}M

Q54 The stereoisomers that are formed by electrophilic addition of bromine to $trans - but - 2 - ene$ is/are:

(1) 2 identical mesomers
(2) 2 enantiomers
(3) 1 racemic and 2 enantiomers

(4) 2 enantiomers and 2 mesomers

Q55 According to Heisenberg's uncertainty principle, the product of uncertainty in position and velocity for an electron is
[Given that mass of electron $= 9.1 \times 10^{-31}$ kg Planck's constant $= 6.63 \times 10^{-34}$ kg m^2 s^{-1}]
(1) 2.8×10^{-3} m^2 s^{-1}
(2) 6.8×10^{-6} m^2 s^{-1}
(3) 5.8×10^{-5} m^2 s^{-1}
(4) 2.8×10^{-5} m^2 s^{-1}

Q56 Find moles of Sn^{2+} which will react with 5 L of $2\,M\,K_2Cr_2O_7$ in acidic medium
(1) 25 (2) 20
(3) 30 (4) 10

Q57 Select the **incorrect** statement from the following options.
I. Meso compound possess both plane of symmetry and centre of symmetry.
II. Meso compound possess either plane or centre of symmetry.
III. Meso compound does not possess either plane or centre of symmetry.
IV. Meso compounds are externally compensated form.
(1) Only I
(2) Only II
(3) I, III and IV only
(4) I, II and III only

Q58 Reaction of cyclohexene with Baeyer's reagent gives:
(1) Cyclohexanol
(2) Cyclohexanone
(3) Cis-cyclohexane-1,2-diol
(4) Trans-cyclohexane-1,2-diol

Q59 The value of n-factor in the following disproportionation reaction is:
$Br_2 \rightarrow Br^- + BrO_3^-$
(1) 1/2 (2) 2/3
(3) 5/3 (4) 7/3

Q60 Which of the following are isoelectronic and isostructural?

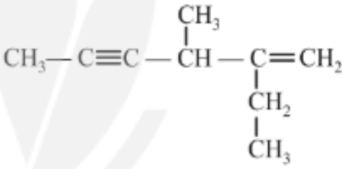
$$NO_3^-, CO_3^{2-}, ClO_3^-, SO_3$$

(1) NO_3^-, CO_3^{2-}
(2) SO_3, NO_3^-
(3) ClO_3^-, CO_3^{2-}
(4) CO_3^{2-}, ClO_3^-

Q61 The density of a 3 M NaCl solution is 1.252 g/ml. Calculate the molality of the solution:
(1) 2.18 m (2) 3.00 m
(3) 2.60 m (4) 2.79 m

Q62 Two litres of an ideal gas at a pressure of 10 atm expands isothermally into a vacuum until its total volume is 10 litres. How much heat is absorbed and how much work is done in the expansion?
(1) q = 100J, w = -100J
(2) q = 0, w = 0
(3) q = 0, w = -100J
(4) q = 100J, w = 0

Q63 The IUPAC name of the given compound is:



(1) 2-Ethyl-3-methyl-hexa-1-en-4-yne
(2) 2-Ethyl-3-methyl-hexa-1-en-4-yne
(3) 3-Methylidene-4-methylhepta-5-yne
(4) 4-Methyl-5-methylidenehepta-2-yne

Q64 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The maximum number of electrons in a subshell that can have the quantum numbers, n = 4, *l* = 1 are 6.
Reason R: p-subshell can have maximum of 6 electrons.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.

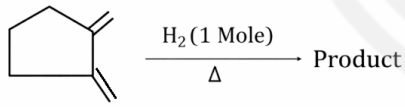
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

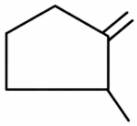
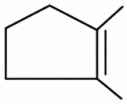
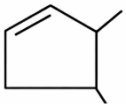
Q65 2 mol of N_2 is mixed with 6 mol of H_2 in a closed vessel of one litre capacity. If 50% of N_2 is converted into NH_3 at equilibrium, the value of K_c for the reaction.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is
(1) 4/27 (2) 27/4
(3) 1/27 (4) 24

Q66 For the reaction,
$2\,NO_2\,(g) + \frac{1}{2}O_2(g) \rightleftharpoons N_2O_5\,(g)$
if the equilibrium constant is K_p, then the equilibrium constant for the reaction
$2N_2O_5\,(g) \rightleftharpoons 4\,NO_2\,(g) + O_2\,(g)$ would be:
(1) K_p^2 (2) $\frac{2}{K_p}$
(3) $\frac{1}{K_p^2}$ (4) $\frac{1}{\sqrt{K_p}}$

Q67 Molarity of an aqueous solution of H_2SO_4 is 18M. If the density of solution is 1.8 g/cm^3, then the molality of solution is:
(Molar mass of H_2SO_4 = 98 g/mol)
(1) 180 m (2) 250 m
(3) 360 m (4) 500 m

Q68



(1) 
(2) 
(3) 
(4) None of these

Q69 Zinc and hydrochloric acid react according to the reaction.
$Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq.) + H_2(g)$
If 0.30 mole of Zn are added to hydrochloric acid containing 0.52 mole HCl, how many moles of H_2 are produced?
(1) 0.26 (2) 1.04
(3) 0.52 (4) 0.13

Q70



Comment upon optical activity of the product.
(1) Racemic mixture
(2) Diastereomers
(3) Meso
(4) Optically inactive due to absence of chiral center

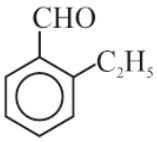
Q71 A reaction has $\Delta H = -33$ kJ mol^{-1} and $\Delta S = -58$ J/K mol. This reaction would be:
(1) spontaneous at all temperatures
(2) non-spontaneous at all temperatures
(3) spontaneous above a certain temperature only
(4) spontaneous below a certain temperature only

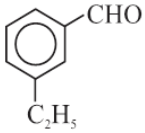
Q72 What is the potential of a half-cell consisting of zinc electrode in 0.01 M $ZnSO_4$ solution at 25°C ?
($E^\circ_{O.P.}$ = 0.763 V)
(1) 0.8221 V (2) 8.221 V
(3) 0.5282 V (4) 9.232 V

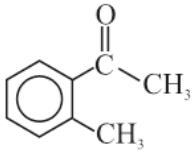
Q73 The decreasing order of boiling points of the following alcohols is
I. 1-pentanol
II. 2-pentanol
III. 3-methyl-2-butanol
IV. 2-methyl-2-butanol.
(1) I > II > III > IV
(2) I > III > II > IV
(3) II > I > III > IV
(4) IV > III > II > I

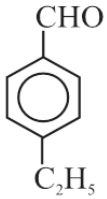
Q74 An aromatic compound 'X' with molecular formula $C_9H_{10}O$ gives the following chemical tests
(i) Forms 2,4-DNP derivative

(ii) Reduce Tollen's reagent
(iii) Undergoes Cannizzaro reaction
(iv) On vigorous oxidation, 1,2-benzene dicarboxylic acid is obtained.
The compound (X) is:

(1) 

(2) 

(3) 

(4) 

Q75 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Lucas reagent is a mixture anhydrous $ZnCl_2$ and concentrated HCl.
Reason R: Primary alcohol produces precipitate with Lucas reagent immediately.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

Q76 The d-electron configurations of $Cr^{2+}, Mn^{2+}, Fe^{2+}$ and Co^{2+} are d^4, d^5, d^6 and d^7 respectively.
Which one of the following will exhibit minimum paramagnetic behaviour?
(1) $[Mn(H_2O)_6]^{2+}$ (2) $[Fe(H_2O)_6]^{2+}$
(3) $[Co(H_2O)_6]^{2+}$ (4) $[Cr(H_2O)_6]^{2+}$

Q77 Given below are two statements

Statement I: S_N1 reaction proceeds with racemization.
Statement II: In S_N2 reactions, retention of configuration take place for optically active reactant
In the light of the above statements, choose the *most appropriate* answer from the options given below
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

Q78 Match the List-I with List-II:

List-I	List-II
(A) Benzenesulphonyl chloride	(I) Test for primary amines
(B) Hoffmann bromamide reaction	(II) Anti Saytzeff
(C) Carbylamine reaction	(III) Hinsberg reagent
(D) Hoffmann orientation	(IV) Known reaction of Isocyanates

Choose the correct answer from the options given below:

[JEE Main 2024, 8 Apr (Shift 1)]
(1) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
(2) (A) - (IV), (B) - (II), (C) - (I), (D) - (III)
(3) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
(4) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Q79 In the following reaction sequence in aqueous solution, the species X, Y and Z, respectively, are

$$S_2O_3^{2-} \xrightarrow{Ag^+} X\,[\text{Clear solution}] \xrightarrow{Ag^+} Y$$
$$[\text{White precipitate}] \xrightarrow{\text{with time}} Z$$
$$[\text{Black precipitate}]$$

(1) $[Ag(S_2O_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag_2S
(2) $[Ag(S_2O_3)_3]^{5-}$, $Ag_3S_2O_3$, Ag_2S
(3) $[Ag(S_2O_3)_2]^{3-}$, $Ag_2S_2O_3$, Ag
(4) $[Ag(SO_3)_3]^{3-}$, Ag_2SO_4, Ag

Q80

White precipitate of $AgCl$ turns to greyish or black when it:
(1) reacts with Na_3AsO_3.
(2) is exposed to sunlight.
(3) reacts with K_2CrO_4.
(4) reacts with concentrated HCl.

Q81 Which one of the following complexes shows optical isomerism?
(1) $cis-[PtCl_2(en)_2]^{2+}$
(2) $trans\ -[PtCl_2(en)_2]^{2+}$
(3) $[CoCl_2(NH_3)_4]Cl$
(4) $trans-[CrCl_2(ox)_2]^{3-}$

Q82 Consider the following statements:
1. $NaBH_4$ can reduce aldehydes and ketones but not carboxylic acids.
2. $LiAlH_4$ can reduce carboxylic acids and esters to alcohols.
3. Catalytic hydrogenation (H_2/Ni) reduces $C=C$ double bonds faster than $C=O$.
4. $LiAlH_4$ reacts violently with water, but $NaBH_4$ is comparatively safer.
Which of the above statements are correct?
(1) 1 and 2 only
(2) 1, 2 and 4 only
(3) 2, 3 and 4 only
(4) All of these

Q83

Statement I: E_{cell} is an intensive parameters.
Statement II: The SI unit of resistivity is ohm^{-1} m.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q84 The IUPAC name of the complex $[Co(NH_3)_4(H_2O)Cl]Cl_2$ is:
(1) Aquatetramminechloridocobalt (III) chloride
(2) Chloridoaquatetramminecobalt (III) chloride
(3) Chloridoaquatetramminecobalt (II) chloride
(4) Tetraammineaquachloridocobalt (III) chloride

Q85 $Fe(OH)_3$ and $Cr(OH)_3$ precipitates can be completely separated by:
(1) Aq. NH_3 (2) HCl
(3) $NaOH/H_2O_2$ (4) H_2SO_4

Q86 Which one of the following reactions will not form acetaldehyde?
[JEE Mains, 25 Feb (Shift 1)]
(1) $CH_2=CH_2+O_2 \xrightarrow{Pd(II)/\ Cu(II)}$
(2) $CH_3CH_2OH \xrightarrow[573\ K]{Cu}$
(3) $CH_3CH_2OH \xrightarrow{CrO_3-H_2SO_4}$
(4) $CH_3CN \xrightarrow[(ii)\ H_2O]{(i)\ DiBAL-H}$

Q87 For a reaction 2A→3B, if the rate of formation of B is x mol/L, the rate of consumption of A is:
(1) x (2) $\frac{3x}{2}$
(3) $\frac{2x}{3}$ (4) 3x

Q88 Match the complex ions given in Column I with the colours given in Column II and assign the correct code :

Column I (complex ions)	Column II (colours)
A. $[Co(NH_3)_6]^{3+}$	1. Violet
B. $[Ti(H_2O)_6]^{3+}$	2. Green
C. $[Ni(H_2O)_6]^{2+}$	3. Light blue
D. $[Ni(H_2O)_4(en)]^{2+}$ (aq)	4. Yellow-orange
	5. Blue

Code :
(1) A − (1) B − (2) C − (4) D − (5)
(2) A − (4) B − (3) C − (2) D − (1)
(3) A − (3) B − (2) C − (4) D − (1)
(4) A − (4) B − (1) C − (2) D − (3)

Q89 Calculate K_a of acetic acid if its 0.05 M solution has molar conductivity of 7.814 × $10^{-4}\ \Omega^{-1}m^2\ mol^{-1}$ at 25°C.
Given: Λ°_m for $CH_3COOH = 3.907 \times 10^{-2}\ \Omega^{-1}m^2\ mol^{-1}$
(1) $2\ \times\ 10^{-5}$

(2) 1.8×10^{-5}
(3) 4×10^{-4}
(4) 0.02

Q90 A group 15 element forms $d\pi - d\pi$ bond with transition metals. It also forms hydride, which is a strongest base among the hydrides of other group members that form $d\pi - d\pi$ bond. The atomic number of the element is _____

[JEE Main 2025, 28 Jan (Shift 2)]

(1) 13 (2) 14
(3) 15 (4) 16

Q91 Study the given statements carefully and find the option having only the correct ones.
(A) Bacteria, Mycoplasma and cyanobacteria are prokaryotes
(B) Slime moulds, Euglenoids and *Physarum* are examples of unicellular eukaryotes
(C) Mushrooms, toad stool, rust and smuts all are sac fungi
(D) Virus, viroid and lichens all are acellular organisms
(1) A, B (2) B, C
(3) C, D (4) A, D

Q92 In which of the following nucellar cells surrounding the embryosac start dividing. protrude into the embryo sac and develop in to the adventive embryos ?
(1) Citrus and Mango
(2) Parthenium and Grasses
(3) Mango and Parthenium
(4) Citrus and Parthenium

Q93 Which of the following is correct statement:
(1) A typical nucleosome contains 400 bp of DNA helix.
(2) Lysine and arginine carry positive charge in their side chains.
(3) Nucleosome in chromatin are seen as beads on string structure when view under light microscope
(4) Length of DNA in a haploid cell of human is 2.2 meter.

Q94 Read the following statement and select the correct option :-
(A) Loosely linked gene show high recombination
(B) genes show linkage present on non homologous chromosomes
(C) In garden pea plant 14 linkage groups are present
(D) Linkage is broken by crossing over
(1) A and C (2) A, C and D
(3) B and D (4) A and D

Q95 How many of the following enzymes / proteins are used during DNA Replication?
(A) DNA polymerase
(B) β- galactosidase
(C) DNA ligase
(D) Topoisomerase
(E) Permease
(F) Peptidyl transferase
(1) Three (2) Four
(3) Two (4) Five

Q96 Simple fruit is found in :-
(1) Liliaceae
(2) Solanaceae
(3) Fabaceae
(4) All of the above

Q97 Find the incorrect combination from the four options :-
(1) Lycopsida - Selaginella, Lycopodium
(2) Pteropsida - Dryopteris, Pteris, Adiantum
(3) Bryopsida(moss)- Funaria, Polytrichum Sphagnum
(4) Phaeophyceae - Volvox, Fucus, Spirogyra

Q98 In which of the following the ratio of genotypic and phenotypic are same?
(1) Co - dominance
(2) Incomplete dominance
(3) Both of these
(4) None of them

Q99 Harmful mutations persist in the gene pool because:
(1)

They are recessive and carried by homozygous individuals
(2) They are recessive and carried by heterozygous individuals
(3) They are formed repeatedly
(4) They undergo genetic drift

Q100 Match **List-I** with **List-II**.

List-I		**List-II**	
A.	Sterile stamen	**I.**	Staminode
B.	Stamens attached to petals	**II.**	Epiphyllous
C.	Stamens attached to perianth	**III.**	Epipetalous
D.	Free stamens	**IV.**	Polyandrous

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-II, D-IV
(3) A-III, B-I, C-IV, D-II
(4) A-IV, B-III, C-I, D-II

Q101 Mark the following statements as true (T) or false (F) and select the correct option.
I. In mustard, filaments of stamens are of similar lengths.
II. In Cruciferae, the outer two stamens are short, and the inner four stamens are long.
III. The fruit of the china rose is a capsule.
IV. Marigold, sunflower, and *Chrysanthemum* are members of the family Malvaceae.

	I	II	III	IV
(*a*)	False	True	True	False
(*b*)	True	False	True	False
(*c*)	False	False	False	True
(*d*)	True	True	False	True

(1) a (2) b
(3) c (4) d

Q102 Assertion: In most eukaryotic cell, the net gain of ATP per glucose molecule oxidized is 38 ATP.
Reason: Only three molecules of ATP are synthesised during glycolysis.
(1) Both assertion and reason are correct and reason is correct explanation of assertion
(2) Both assertion and reason are correct and reason is not correct explanation of assertion
(3) Only assertion is true but reason is false
(4) Both assertion and reason are false

Q103 Which of the following biomolecules is common to respiration mediated breakdown of fats, carbohydrates and proteins
(1) Pyruvic acid
(2) Acetyl CoA
(3) Glucose-6-phosphate
(4) Fructose 1,6-bisphosphate

Q104 Which enzyme of the Krebs cycle is directly linked to the electron transport chain as part of Complex II?
(1) Citrate synthase
(2) Isocitrate dehydrogenase
(3) Succinate dehydrogenase
(4) Malate dehydrogenase

Q105 Choose the **incorrectly** matched pair.
(1) Abingdon tortoise & goat in Galapagos Islands - Mutualism
(2) *Chathamalus* and *Balanus* - Competition
(3) *Cuscuta* on hedge plants - Parasitism
(4) Pisaster - Predator

Q106 **Statement I**: Interspecific interactions arise from the interaction of populations of two different species.
Statement II: Commensalism is the interaction in which one species benefits and the other is neither harmed nor benefited.
(1) Statement I and Statement II both are correct.
(2) Statement I is correct, but Statement II is incorrect.
(3) Statement I is incorrect, but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

Q107 According to logistic growth model, when does a population reach its carrying capacity (K)?
(1) When resources are unlimited
(2) When population density is zero
(3) When competition for limited resources slows down growth
(4) When natality exceeds mortality indefinitely

Q108 Which of the following statement is **incorrect** regarding fungi?
(1) Wheat rust is caused by *Puccinia*.
(2) Sexual reproduction is by oospores, ascospores and basidiospores.
(3) Fungi prefer to grow in cool and humid places.
(4) With the exception of yeasts, fungi are filamentous.

Q109 Given below are two statements:
Statement I: No virus contains both RNA and DNA.
Statement II: A virus is a nucleoprotein and the genetic material is non-infectious.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q110 Given below are two statements:
Statement I: The ethical argument for conserving biodiversity relates to what we owe to millions of plant, animal and microbe species with whom we share this planet.
Statement II: Philosophically or spiritually, we need to realise that every species has an intrinsic value, even if it may not be of current or any economic value to us.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q111 One of the ex situ conservation methods for endangered species is
(1) Wild life sanctuaries
(2) Bioshphere reserves
(3) Cryopreservation
(4) National Park

Q112 Which of the following characters make species more susceptible to extinction?
(1) Ability of species to switch over to alternate foods.
(2) Species restricted to small geographical area.
(3) Small body size and high reproductive potential.
(4) None of these

Q113 Mark the incorrect statement.
(1) In leptotene, the chromosomes become gradually visible under the light microscope.
(2) Bivalents become clearly visible in Zygotene.
(3) Prophase I is typically longer and more complex as compared to prophase of mitosis.
(4) Number of tetrads are equal to the number of bivalents.

Q114 A bivalent of meiosis I consists of _____.
(1) two chromatids and one centromere
(2) two chromatids and two centromeres
(3) four chromatids and two centromeres
(4) four chromatids and four centromeres

Q115 Identify the **incorrect** statement from the following.
(1) Plant growth is generally indeterminate.
(2) Meristems have the capacity to divide and self-perpetuate.
(3) In dicotyledonous plants and gymnosperms, the lateral meristems present from the

beginning.
(4) Plants retain the capacity for unlimited growth throughout their life.

Q116 Absolute growth rate is defined as the:
(1) synthesis of new intercellular and extracellular materials.
(2) measurement and the comparison of total growth per unit time.
(3) growth of the given system per unit time.
(4) increased growth per unit time.

Q117 Match List-I with List-II.

List-I		List-II	
(A)	Growth	(I)	Seen only in humans
(B)	Reproduction	(II)	Defining feature of all living organisms
(C)	Self-consciousness	(III)	Shown by some non-living objects also
(D)	Metabolism	(IV)	Does not take place in mules

Choose the correct answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-IV, C-I, D-II

Q118 Which of the following statement is NOT correct?
(1) Biologists follow universally accepted principles to provide scientific names to known organisms.
(2) Each name has two components - the Generic name and the specific epithet.
(3) This system of providing a name with two components is called Binomial nomenclature.
(4) The scientific name of mango is written as *Mangifera Indica.*

Q119 Read the following statements.
A. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell.
B. No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for lysosome.
C. Mesosome is the extension of cell wall into the cell.
D. The eukaryotes include all the protists, plants, animals and fungi.
How many of the above statements is/are **correct**?
(1) One (2) Two
(3) Three (4) Four

Q120 Plasmodesmata are:
(1) Channels for water passage in the cell membrane
(2) Infoldings of the cell membrane
(3) Cytoplasmic bridges between neighboring cells
(4) Pores in the secondary cell wall

Q121 Select the **correct** option for **true (T)** and **false (F)** for cell membrane.

1. The fluid nature of the membrane is important for endocytosis.
2. Neutral solutes may move across the membrane by the process of simple diffusion against the concentration gradient.
3. The quasi-fluid nature of lipid enables parallel movement of proteins within the overall bilayer.

	I	II	III
(1)	F	T	F
(2)	T	T	F
(3)	T	F	F
(4)	F	T	T

(1) 1 (2) 2
(3) 3 (4) 4

Q122 Find the false statement among the followings.
(1) The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter.
(2) By the process of leaching, water soluble inorganic nutrients move upward on the soil horizon and get dissolved as available salts.
(3) In a terrestrial ecosystem, major producers are herbaceous and woody plants.
(4)

The saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

Q123 How many statements are correct?
(a) Earthworm being referred to as the female's 'friend'.
(b) Earthworms help in the breakdown of complex organic matter.
(c) Earthworms help in loosening the soil.
(d) Earthworms enhance soil properties like moisture retention, nutritional content, and water holding ability.
Select the correct option:
(1) One (2) Two
(3) Three (4) Four

Q124 Stratification in an ecosystem refers to:
A. The vertical distribution of species at different levels.
B. Trees occupying the top vertical strata or layer of a forest.
C. Herbs occupy the second layer.
D. Grasses occupying the bottom layers.
(1) Only A, B, and D are correct
(2) Only B, C, and D are correct
(3) Only A and D are correct
(4) All statements are correct

Q125 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Bottled fruit juices bought from the market are clearer as compared to those made at home.
Reason R: Bottled fruit juices are clarified by the use of lipases.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is not the correct explanation of A.

Q126 Biofertilisers include
(1) nitrogen-fixing cyanobacteria only
(2) nitrogen-fixiing bacteria only
(3) protistans only
(4) bacteria, cyanobacteria and fungi

Q127 *Monascus purpureus*, a yeast, used commercially in the production of
(1) Ethanol
(2) Streptokinase or removing clots from the blood vessels
(3) Citric acid
(4) Blood cholesterol lowering statins

Q128 The major pigments in green algae are ___**I**___ and ___**II**____ and stored food is ___**III**___. Identify **I**, **II** and **III** respectively.
(1) I-chl *a,* II-chl *d,* III-starch
(2) I-chl *a,* II-chl *c,* III-floridean starch
(3) I-chl *a,* II-chl *b,* III-starch
(4) I-chl *a,* II-chl *b,* III-mannitol

Q129 Choose the incorrect statement about mosses.
(1) The fusion of antheridia and archegonia produced at the apex of the leafy shoots leads to sexual reproduction
(2) The sporophyte in mosses is more elaborate than that in liverworts
(3) Seta and capsule bear spores, which give rise to gametophyte after meiosis
(4) Sporophyte is differentiated into foot, seta and capsule

Q130 Which of the following is a mismatched pair?
(1) Storage of pollen grains: $-196^{\circ}\mathrm{C}$
(2) Pollen allergy: Carrot grass
(3) Chasmogamous flowers: Exposed anthers and stigmas
(4) Xenogamy: Self-pollination

Q131 Choose the incorrect match.
(1) Vessels - Cells are dead with thick chitinous cell walls
(2) Tracheids - Elongated tube-like cells with-thick, lignified walls and tapering ends
(3)

Xylem fibres - Cells possess highly thickened walls with obliterated central lumen.
(4) Xylem parenchyma-Cells are living with thin cellulosic cell walls

Q132 Which is of the following is true about metaphloem?
(1) Lack of phloem fibres
(2) Presence of sieve tube
(3) Not formed from vascular cambium
(4) All of the above

Q133 Grafting is not possible in monocots because they
(1) Lack cambium
(2) Have scattered vascular bundles
(3) Have parallel venation
(4) Are herbaceous

Q134 Which of the following statements is/are not correct regarding chlorophyll a?
(i) Chlorophyll b, xanthophylls and carotenoids absorb light and transfer the energy to chlorophyll a.
(ii) It is the primary photosynthetic pigment.
(iii) In pure state, it is red in colour and thus it absorbs more blue wavelength of light than the red wavelength.
(iv) Most of the photosynthesis takes place in the green regions of the spectrum.
(1) (i) and (iii) (2) (iii) and (iv)
(3) (ii) only (4) (iv) only

Q135 The plants with Hatch and Slack pathway are adapted to
(1) humid temperate regions
(2) dry temperate regions
(3) humid tropical regions
(4) dry tropical regions

Q136 Liver and pancreas of frog are derived from
(1) ectoderm
(2) mesoderm
(3) endoderm
(4) Both (b) and (c)

Q137 Which is incorrect related to the part of tubule lined by simple cuboidal brush border epithelium?
(1) Selective secretion of hydrogen ions, ammonia and potassium ions into the filtrate.
(2) absorption of HCO3– from filtrate.
(3) Nearly all of the essential nutrients, and 99 per cent of electrolytes and water are reabsorbed by this segment.
(4) Maintain the pH and ionic balance of the body fluids.

Q138 Skeletal muscle fiber has light and dark bands. Which is the correct match of protein with its light refractive property and colour?
(1) Myosin Light Anisotropic
(2) Actin Dark Anisotropic
(3) Myosin Dark Isotropic
(4) Actin Light Isotropic

Q139 Match the **List-I** with **List-II**.

List-I (Organisms)		**List-II (Respiratory organisms)**	
(A)	Shark	(I)	Lungs
(B)	Insects	(II)	Moist skin
(C)	Bat	(III)	A network of tracheal tubes
(D)	Earthworm	(IV)	Gills

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-I, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-II, D-I

Q140 Which of the following .hormones can play a significant role in osteoporosis?
(1) Aldosterone and Prolactin
(2) Parathyroid hormone and Prolactin
(3) Estrogen and Parathyroid hormone
(4) Progesterone and Aldosterone

Q141 How is NaCl transported in the renal system?
(1)

NaCl is transported by the descending limb of Henle's loop and exchanged with the ascending limb of vasa recta.
(2) NaCl is transported by the ascending limb of Henle's loop and exchanged with the descending limb of vasa recta.
(3) NaCl is transported by the collecting tubule and exchanged with the thin segment of the ascending limb of Henle's loop.
(4) NaCl is transported by the ascending limb of vasa recta and exchanged with the descending portion of Henle's loop.

Q142 Which of the following accurately represents the proportion of oxygen transported in plasma?
(1) Approximately equal to the percentage of neutrophils among total WBCs.
(2) Higher than the percentage of lymphocytes among total WBCs.
(3) Lower than the percentage of basophils among total WBCs.
(4) Approximately equal to the percentage of eosinophils among total WBCs.

Q143 Thrombokinase is associated with
(1) Production of erythrocytes from the bone marrow
(2) Pulmonary and systemic circulation
(3) Cardiac cycle and its regulation
(4) Enzymatic reactions in coagulation of blood

Q144 The signal transduction of steroid hormone across cell is through
(1) binding of hormone to the cytoplasmic receptor and the complex binds to hormone response element on DNA within promoter DNA
(2) binding of hormone to the transmembrane receptor which initiates the production of second messenger that activates enzymes which further activates transcription factors
(3) binding of hormone to the transmembrane receptor which diffuse inside the cell cytoplasm and then activates the enzyme necessary for the activation of transcription factors
(4) binding of hormone to the cytoplasmic receptor that initiates the production of second messenger which activates enzymes that further activates transcription factors.

Q145 Select the correct option with respect to cockroaches:
(1) Malpighian tubules convert nitrogenous wastes into urea
(2) Males bear short anal styles not present in females
(3) Nervous system comprises of a dorsal nerve cord and ten pairs of ganglia
(4) The forewings are tegmina which are used in flight

Q146 Match **List-**I with **List-II**:

List-I		**List-II**	
(A)	6-15 pairs of gill slits	(I)	*Corvus*
(B)	Gill slits are without operculum.	(II)	Osteichthyes
(C)	Ctenoid scales	(III)	Chondrichthyes
(D)	Pneumatic bones	(IV)	*Myxine*

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-IV, B-III, C-II, D-I

Q147 A patient suffering from severe dehydration is likely to exhibit which of the following physiological responses?
(1) Decreased release of ADH, leading to increased urine output.
(2) Increased blood volume, leading to inhibition of ADH release.
(3) Suppression of osmoreceptors, leading to decreased blood pressure and GFR.
(4) Activation of osmoreceptors, leading to increased ADH secretion and water reabsorption.

Q148

When a person is suffering from poor renal reabsorption, which one of the following will not help in maintenance of blood volume?
(1) increased A DH secretion.
(2) decreased glomerular filtration.
(3) increased arterial pressure in kidneys.
(4) decreased arterial pressure in kidneys.

Q149 Cellulose is made of
(1) Unbranched chain of glucose molecules linked by $\alpha - 1,6$ glycosidic bonds
(2) Unbranched chain of glucose molecules linked by $\beta - 1,4$ glycosidic bonds
(3) Branched chain of glucose molecules having $\alpha - 1,6$ glycosidic bonds at the site of branching
(4) Branched chain of glucose molecules with $\alpha - 1,6$ glycosidic bonds in the straight chain and $\beta - 1,4$ linked bonds at the site of branching

Q150 Study the following statements:
I. Low temperature preserves the enzyme in a temporarily inactive state.
II. Enzymes isolated from thermophilic organisms get denatured at 50 °C.
III. The active site of enzyme breaks the chemical bonds of the product.
IV. Prosthetic groups are tightly bound to the apoenzyme.
Select the option with **correct** statements:
(1) I and III only
(2) III and IV only
(3) II and III only
(4) I and IV only

Q151 Match **List-I** with **List-II**:

List-I		List-II	
(A)	Cockroach	(I)	Flame cells
(B)	Earthworm	(II)	Antennal glands
(C)	*Planaria*	(III)	Malpighian tubules
(D)	Prawn	(IV)	Nephridia

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-IV, C-III, D-I

Q152 Read the following statements (**I-IV**).
I. Human circulatory system consists of a muscular chambered heart, blood vessels and blood.
II. Heart is mesodermal in origin.
III. The walls of ventricles are much thicker than that of the atria.
IV. In birds oxygenated and deoxygenated blood is received by the left and right atria respectively.
Which of the above statements are **correct**?
(1) I and III only
(2) I, II and IV only
(3) II, III and IV only
(4) I, II, III and IV

Q153 Which of the following properties of starch makes it useful as storage material?
A. Easily translocated
B. Chemically non-reactive
C. Easily digested by animals
D. Osmotically inactive
E. Synthesized during photosynthesis
(1) A, C & E (2) A and E
(3) B and C (4) B and D

Q154 Common characteristics of cockroach, housefly and mosquito are
(1) compound and simple eyes
(2) one pair of wings and halteres
(3) three pairs of legs and one pair of antennae
(4) two pairs of legs and two compound eyes

Q155 Given below are two statements.
Statement I: An adrenal medullary hormone can increase cardiac output.
Statement II: RBCs have an average life span of 120 days.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1)

Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q156 Select the **correct** option to complete the analogy w.r.t hormone and its effect/function.
Catecholamine : Pupillary dilation , ____: : Reabsorption of Na^+ and water
(1) Aldosterone (2) Insulin
(3) Epinephrine (4) Glucagon

Q157 Which one of the following pairs of structures is correctly matched with their correct description?

	Structure	Description
A	Tibia and Fibula	Both form parts of knee joint
B	Cartilage and cornea	No blood supply but do require oxygen for respiratory need
C	Shoulder and elbow joint	Ball and socket joint
D	Thigh bone (femur)	Articulates with the glenoid cavity

(1) A (2) B
(3) C (4) D

Q158 Which of the following hormone is not chemically glycoprotein?
(1) Growth hormone
(2) Prolactin
(3) Luteinizing hormone
(4) Estrogen

Q159 Cerebral malignant malaria is caused by
(1) *Plasmodium ovale*
(2) *Plasmodium vivax*
(3) *Plasmodium falciparum*
(4) *Plasmodium malariae*

Q160 During proliferative phase, uterine wall undergoes certain changes, these are
(1) Myometrium wall is sloughed off
(2) Endometrium wall is sloughed off
(3) Blood vessel in endometrium becomes long and coiled
(4) Proliferation of myometrial epithelial living

Q161 Read the following statements and choose the option with **correct** statements.
I. Implants are used under skin.
II. Multiload 375 consists of six matchstick sized tubes containing hormones.
III. Mode of action of implants is similar to that of pills.
IV. Effective period of implants is shorter than pills.
V. Emergency contraceptives could be used to avoid possible pregnancy due to rape.
(1) I,II and III only
(2) I,III, IV and V
(3) II, IV and V only
(4) I and II only

Q162 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion (A): Not all copulations lead to pregnancy.
Reason (R): Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampulla region.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Assertion (A) and Reason (R) are the true, and Reason (R) is a correct explanation of Assertion (A).
(2) Both Assertion (A) and Reason (R) are the true, and Reason (R) is not a correct explanation of Assertion (A).
(3) Assertion (A) is true, and Reason (R) is false.
(4) Assertion (A) is false, and Reason (R) is true.

Q163 Read the following statements and state **True(T)** or **False (F)**.
I. Each ovary is covered by a thin epithelium which encloses the ovarian stroma.
II. The ovarian stroma is divided into two zones – a peripheral medulla and an inner cortex.
III. The myometrium undergoes cyclical changes during menstrual cycle.

IV. The male reproductive system is located in the pelvis region.
(1) I-T; II-F; III-F; IV-T
(2) I-T; II-F; III-T; IV-F
(3) I-F; II-T; III-F; IV-F
(4) I-F; II-T; III-T; IV-F

Q164 Trace the correct path of movement of the sperms upto urethra.
(1) Seminiferous tubules → Vasa efferentia → Rete testis → Epididymis → Vas deferens → Ejaculatory duct → Urethra
(2) Seminiferous tubules → Rete testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Urethra
(3) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra
(4) Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Ejaculatory duct → Vas deferens → Urethra

Q165 Which of the following options is **correct** regarding the sex determination method in human beings?
(1) The sex of offspring is determined by environmental factors.
(2) The sex of a child is determined by the type of sperm that fertilises the egg.
(3) The mother's chromosomes solely determine the sex of the child.
(4) The presence of X chromosomes in the egg determines the sex of the child.

Q166 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Green revolution was comparatively less effective in developing world where farmers were dependent on conventional breeding.
Reason: In developing world, inability to buy expensive agro-chemicals forced farmers to rely on conventional breeding.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

Q167 After the transformation of spermatids into sperm, their heads become embedded in a cell called X and are finally released from the Y by the process called Z. Identify X, Y and Z.

	X	Y	Z
1	Spermatogonium	Epididymis	Insemination
2	Leydig	Vas deferens	Parturition
3	Sertoli	Seminiferous tubules	Spermiation
4	Spermatocyte	Seminiferous tubules	Spermiogenesis

(1) 1 (2) 2
(3) 3 (4) 4

Q168 Identify the correct pair.

Proteins	Functions
(a) Collagen	Hormone
(b) Antibody	Fights infectious agents
(c) Insulin	Intercellular ground substance
(d) Trypsin	Enables glucose transport in cell

(1) (a) (2) (b)
(3) (c) (4) (d)

Q169 A GM crop is transgenic. It can be harmful because it may cause
(1) allergies and toxicity.
(2) incorporation of antibiotic resistance in human beings.
(3) disturbance in metabolism due to enzyme for antibiotic resistance.
(4) All of the above

Q170 A paleontologist has recovered a tiny bit of organic material from the 400-year-old preserved skin of an extinct dodo. She would like to compare DNA from the sample with DNA from living birds. Which of the following would be most useful for increasing the amount of DNA available for testing?
(1) restriction fragment analysis
(2) polymerase chain reaction
(3) molecular probe analysis
(4) electrophoresis

Q171 In vitro fertilisation is a technique that involves transfer of which one of the following into the fallopian tube?
(1) Embryo only, upto 8 cell stage
(2) Either zygote or early embryo upto 8 cell stage
(3) Embryo of 32 cell stage
(4) Zygote only

Q172 Diacetyl morphine is commonly known as
(1) cocaine (2) hashish
(3) ganja (4) heroin

Q173 Intense lactation in mothers acts as a natural contraceptive due to the:
(1) Suppression of gonadotropins
(2) Hypersecretion of gonadotropins
(3) Suppression of gametic transport
(4) Suppression of fertilization

Q174 Read the following statements and choose incorrect one.
(1) DNA is hydrophobic molecule, it cannot pass through cell membrane
(2) By micro-injection, recombinant DNA is directly injected into the nucleus of an animal cell
(3) During DNA isolation we used chilled ethanol in final step
(4) Retroviruses have been use to deliver desirable gene into animal cells after disarming

Q175 *Homo erectus* -
A. had a large brain around 900 c. c.
B. probably ate meat.
C. appeared about 1.5 mya year ago
D. evolved from *H. habilis.*
(1) A and B
(2) B and C
(3) None of these are correct
(4) All

Q176 Which of the following drug gets partly converted to morphine in the body?
(1) Mescaline (2) Benzedrine
(3) Psilocybin (4) Codeine

Q177 Consider the following four statements (i - iv) regarding kidney transplant and select the two correct ones out of these.
(i) Even if a kidney transplant is proper, the recipient may need to take immuno-suppressants for a long time.
(ii) The cell-mediated immune response is responsible for the graft rejection.
(iii) The B-lymphocytes are responsible for rejection of the graft.
(iv) The acceptance or rejection of a kidney transplant depends on specific interferons.
The two correct statements are
(1) (ii) and (iii) (2) (iii) and (iv)
(3) (i) and (iii) (4) (i) and (ii)

Q178 Assertion (A) : Both male and female can be infertile.
Reason (R) : Infertile couples can have children with the help of ARTs.
Choose the right answer from the options given below
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

Q179 An athlete is showing characters like increased aggressiveness, mood swings, depression, decreased sperm production, breast enlargement indicating;
(1) addiction of alcohol.
(2) addiction for drugs like morphine.
(3) misuse of sedatives to enhance performance.
(4) misuse of anabolic steroids to enhance performance.

Q180 Assertion: The double-stranded DNA is denatured by subjecting it to a high temperature of 94°C for 15 seconds.
Reason: One of the separated strands of DNA is destroyed during denaturation, and the other

acts as a template.

(1) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

(3) Assertion is true, but Reason is false.

(4) Both Assertion and Reason are false.

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TEST-02

NEET AIR (2027)

DATE : 20/08/2026

Answer Key

Q1	(4)
Q2	(2)
Q3	(2)
Q4	(4)
Q5	(4)
Q6	(3)
Q7	(3)
Q8	(4)
Q9	(4)
Q10	(1)
Q11	(2)
Q12	(4)
Q13	(4)
Q14	(2)
Q15	(3)
Q16	(1)
Q17	(3)
Q18	(2)
Q19	(1)
Q20	(3)
Q21	(2)
Q22	(1)
Q23	(4)
Q24	(2)
Q25	(3)
Q26	(2)
Q27	(4)
Q28	(4)
Q29	(3)
Q30	(1)
Q31	(4)
Q32	(2)
Q33	(3)
Q34	(3)
Q35	(2)
Q36	(2)
Q37	(3)
Q38	(2)
Q39	(2)
Q40	(3)
Q41	(4)
Q42	(4)
Q43	(1)
Q44	(3)
Q45	(3)
Q46	(3)
Q47	(4)
Q48	(1)
Q49	(4)
Q50	(4)
Q51	(3)
Q52	(4)
Q53	(4)
Q54	(1)
Q55	(3)
Q56	(3)
Q57	(3)
Q58	(3)
Q59	(3)
Q60	(1)
Q61	(4)
Q62	(2)

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Q63 (1)
Q64 (3)
Q65 (1)
Q66 (3)
Q67 (4)
Q68 (2)
Q69 (1)
Q70 (4)
Q71 (4)
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Q128 (3)


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Q129 (3)
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Q171 (2)
Q172 (4)
Q173 (1)
Q174 (1)
Q175 (4)
Q176 (4)
Q177 (4)
Q178 (2)
Q179 (4)
Q180 (3)


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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

The voltage V_L and V_C are equal and opposite, so voltmeter reading will be zero.

Aslo, $R = 30\ \Omega,\ X_L = X_C = 25\ \Omega$

So, $I = \frac{V}{\sqrt{R^2+(X_L-X_C)^2}} = \frac{V}{R} = \frac{240}{30} = 8\ A$

Video Solution:

Q2 Text Solution:

Due to force of repulsion on the negative charge, velocity decreases and hence the kinetic energy decreases going from B to A .

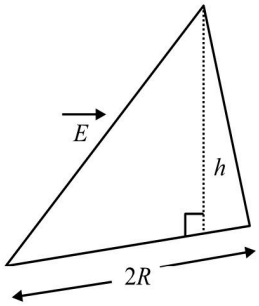
Video Solution:

Q3 Text Solution:

The correct answer is option (2)

The cross section of the cone is a triangle

$\phi = E \times \frac{1}{2} \times 2R \times h = ERh$



Video Solution:

Q4 Text Solution:

$V_T = V_1 + V_2 + V_3 + V_4$

$= \frac{\sigma R}{8\varepsilon_0} + \frac{2\sigma R}{8\varepsilon_0} + \frac{3\sigma R}{8\varepsilon_0} + \frac{4\sigma R}{8\varepsilon_0}$

$= \frac{5\sigma R}{4\varepsilon_0}$

Video Solution:

Q5 Text Solution:

Let the charge on the capacitor C be Q.

Charge on the combination of 1 and 2μF is also Q.

$Q_2 = \frac{2}{1+2}Q = \frac{2}{3}Q$

But, $Q = E\left(\frac{3C}{3+C}\right)$

$\therefore\ Q_2 = \frac{2}{3}E\left(\frac{3C}{3+C}\right) = \frac{2EC}{3+C}$

As we can see, since C is between 1 and 3, Q_2 will increase until $C = 3\ \mu F$

Slope of curve, $\frac{dQ_2}{dC} = \frac{(3+C)2E-2EC}{(3+C)^2} = \frac{6E}{(3+C)^2}$

Thus, slope decreases as (C) increases.

Video Solution:

Q6 Text Solution:

Two wires carry the same current

$I = n_1 eAV_{d_1} = n_2 eAV_{d_2}$

So $n_1 V_{d_1} = n_2 V_{d_2}$

or $\frac{V_{d_1}}{V_{d_2}} = \frac{n_1}{n_2} = 4$

$\Rightarrow V_{d_1} : V_{d_2} = 4 : 1$

Video Solution:

Q7 Text Solution:

Time period of revolution of charged particle in magnetic field is $\frac{2\pi m}{qB}$ which is independent on speed.

Video Solution:

Q8 Text Solution:

First, we find the velocity by differentiating $x(t) = 40 + 12t - t^3$:

$$v(t) = \frac{dx}{dt} = 12 - 3t^2.$$

To find when the particle comes to rest (beyond $t = 0$), set $v(t) = 0$:

$$12 - 3t^2 = 0 \implies t^2 = 4 \implies t = 2\,\text{s}.$$

Next, compute the position at $t = 2$:

$$x(2) = 40 + 12 \times 2 - (2)^3 = 40 + 24 - 8 = 56.$$

Since $x(0) = 40$, the distance traveled from $t = 0$ to $t = 2$ is

$$56 - 40 = 16\,\text{m}.$$

Hence, the particle travels $\boxed{16\,\text{m}}$ before coming to rest.

Video Solution:

Q9 Text Solution:

a = $\frac{14}{4+2+1}$

= $\frac{14}{7}$ = 2 m/s^2

(14 – N) = 4 × 2

N = 14 – 8

= 6 N

Video Solution:

Q10 Text Solution:

For $t < 0\ and\ t > 4\ s$, the position of the particle is not changing i.e. the particle is at rest. So no force is acting on the particle at these intervals.

For $0 < t < 4\ s$, the positions of the particle is continuously changing. As the position-time graph is a straight line, the motion of the particle is uniform, so acceleration, $a = 0$. Hence no force acts on the particle during this interval also.

Video Solution:

Q11 Text Solution:

$\mathrm{P} = \frac{a - t^2}{bx}$

$\mathrm{P} = \left[\mathrm{ML}^{-1}\mathrm{T}^{-2}\right]$

a will have same unit as t^2. So, $a = [\mathrm{T}^2]$

$b = \left[\frac{a - t^2}{\mathrm{P}x}\right] = \frac{[\mathrm{T}^2]}{[\mathrm{ML}^{-1}\mathrm{T}^{-2}\mathrm{L}]} = \left[\mathrm{M}^{-1}\mathrm{L}^0\mathrm{T}^4\right]$

$\frac{a}{b} = \frac{[T^2]}{[M^{-1}L^0T^4]} = [ML^0T^{-2}]$

Video Solution:

Q12 **Text Solution:**

Mass of first block (m_1) = m

Mass of second block (m_2) = $4m$

Initial velocity of first block (u_1) = v

Since second block (heavier black) is stationary initially,

Initial velocity of second block (u_2) = 0

Initial momentum of the system

$(P_1) = m_1u_1 + m_2u_2 = m \times v + 4m \times 0$

$= mv$

After collision, the first block (lighter block) comes to rest,

So final velocity of first block (V_1) = 0

Final velocity of second block = V_2

Final momentum of the system

$(P_2) = m_1v_1 + m_2v_2 = m \times 0 + 4m \times v_2$

$= 4mv_2$

Since there is no external force on the system, so momentum of the system will remain constant

Initial momentum (P_1) = Final momentum (P_2)

$mv = 4m\ v_2$

$v_2 = v/4$

Now use the formula;

$e = \frac{\text{Velocity of separation}}{\text{Velocity of approach}} = \frac{V_2 - V_1}{u_1 - u_2} = \frac{\frac{v}{4} - 0}{v - 0}$

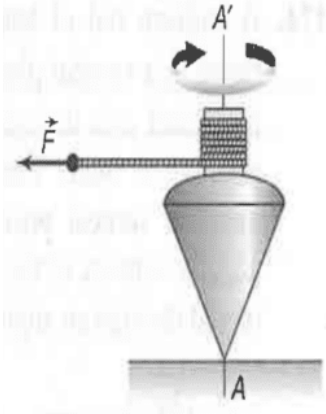
$= \frac{v}{4v} = \frac{1}{4}$

Thus coefficient of restitution (e) =1 / 4=0.25

Video Solution:

Q13 **Text Solution:**

Since the force applied on the string is same as that of the tension in the string.



The work done is given as,

$W = F \times s$

$= 2.5 \times 80 \times 10^{-2}$

$= 2J$

The work energy relation can be written as,

$W = \frac{1}{2}I\omega^2$

$2 = \frac{1}{2} \times 4 \times 10^{-4}\omega^2$

$\omega = 100\frac{rad}{sec}$

Option (4) is the correct answwer.

Video Solution:

Q14 **Text Solution:**

Statement I is correct because the force of gravitation on the planet by sun is central.

Statement II is correct because total mechanical energy of an object is the sum of its K.E. (which is always positive) and the P.E. (relative to infinity is always negative) and since $|KE| < |PE|$, hence total energy is negative.

Statement III is incorrect because p = mv and velocity of planets around sun is not constant at every point.

Video Solution:

Q15 Text Solution:

Initial energy of the asteroid is

$$E_i = K_i + U_i = \frac{1}{2}mv_i^2 - \frac{GM_E\, m}{R_E}$$

Final energy of the asteroid is

$$E_f = \frac{1}{2}\, mv_f^2 - \frac{GM_E\, m}{R_E}$$

According to law of conservation of energy,

$$E_i = E_f$$

$$\frac{1}{2}\, mv_i^2 - \frac{GM_E m}{10R_E} = \frac{1}{2}\, mv_f^2 - \frac{GM_E\, m}{R_E}$$

$$v_f^2 - \frac{2\, GM_E}{R_E} = v_i^2 - \frac{2\, GM_E}{10R_E}$$

$$v_f^2 = v_i^2 + \frac{2\, GmE}{R_E}\left(1 - \frac{1}{10}\right)$$

So, option (3) is correct.

Q16 Text Solution:

Lengths are same $\Rightarrow l_1 = l_2$,

Area of cross sections are same, $A_1 = A_2$

$Y_1 = 0.9 \times 10^{11}$ N/m^2

$Y_2 = 3.60 \times 10^{11}$ N/m^2

Elongation, $e = \frac{Fl}{AY}$

$\therefore \quad \frac{e_1}{e_2} = \frac{F_1 l_1}{A_1 Y_1} \cdot \frac{A_2 Y_2}{F_2 l_2}$

($\because$ F, l and A are same)

$$\therefore \quad \frac{e_1}{e_2} = \frac{3.60 \times 10^{11}}{0.9 \times 10^{11}}$$

$$= \frac{4}{1} \text{ or } e_1 : e_2 = 4 : 1$$

Video Solution:

Q17 Text Solution:

Beat frequency $\frac{v}{2l} - \frac{v}{2(l+x)} = \frac{v(l+x-l)}{2l(l+x)} = \frac{vx}{2l^2}$

Q18 Text Solution:

$$(W_1) = MB(\cos 0^\circ - \cos 90^\circ)$$

$$= MB(1 - 0) = BM$$

Similarly

$$W_2 = MB(\cos 0^\circ - \cos 60^\circ)$$

$$= MB\left(1 - \frac{1}{2}\right) = \frac{MB}{2}$$

Therefore, $W_1 = 2W_2$ or $n = 2$.

Video Solution:

Q19 Text Solution:

I for diamagnetic substance is very small and negative. Hence the graph should be in negative direction with low value OC.

Video Solution:

Q20 Text Solution:

As I increases, ϕ increases

$\therefore$ I_i is such that it opposes the increases in ϕ. Hence, ϕ decreases (By Right Hand Rule). The induced current will be counterclockwise.

Video Solution:

Q21 Text Solution:

$$t_1 = \frac{a}{v}, \; t_2 = \frac{a}{v}$$

$$T = t_1 + t_2$$

$$T = \frac{2a}{v}$$

Video Solution:



Q22 **Text Solution:**

$$\frac{I}{O} = \frac{f}{f-u}$$

$$\Rightarrow \frac{I}{+6} = \frac{-f}{-f-(-4f)}$$

$$\Rightarrow \quad I = -2 \text{ cm}$$

Video Solution:



Q23 **Text Solution:**

E = 13.2 keV

$$E = \frac{12400}{\lambda(\text{Å})}$$

$$\lambda = \frac{12400}{13200}$$

$\lambda \approx$ 1Å (X-Rays)

Video Solution:



Q24 **Text Solution:**

Here, $\text{I'} = \frac{\text{I}}{3}$

According to the law of Malus

$$\text{I'} = \text{I}\cos^2\theta$$

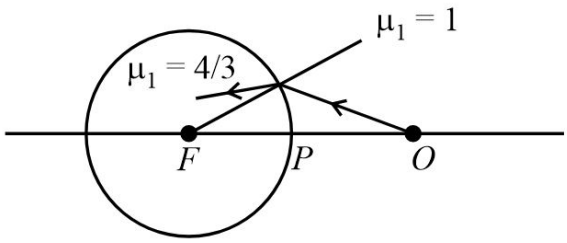
$$\frac{\text{I}}{3} = \text{I}\cos^2\theta$$

$$\cos^2\theta = \frac{1}{3}$$

$$\cos\theta = \frac{1}{\sqrt{3}}$$

$$\theta = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Q25 **Text Solution:**



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2-\mu_1}{R} \Rightarrow \frac{4}{3v} - \frac{1}{-R} = \frac{4/3-1}{R}$$

$$\Rightarrow \frac{4}{3v} = \frac{1}{3R} - \frac{1}{R}$$

$$\frac{4}{3v} = \frac{1-3}{3R} \Rightarrow = -\frac{4R}{2} \Rightarrow v = -2R$$

distance of image from center= R + 2 R = 3 R

Video Solution:



Q26 **Text Solution:**

(2)

$\text{A} \rightarrow \text{III},\ \text{B} \rightarrow \text{IV}, \text{C} \rightarrow \text{II}, \text{D} \rightarrow \text{I}$

Maximum KE is given by

$$K_{\max} = hf - \phi_0$$

$$= \frac{6.62\times10^{-34}\times6\times10^{14}}{1.6\times10^{-19}} - 2.14$$

$$= 2.485 - 2.140 = 0.345\,\text{eV}$$

$K_{\min} = 0,$

Stopping potential,

$$V_0 = \frac{K_{\max}}{e} = \frac{0.345\text{ eV}}{e}$$

Or $V_0 = 0.345\text{ V} = 345\,\text{mV}$

Maximum speed of emitted electrons,

$$K_{\max} = \frac{1}{2}\,\text{m}v_{\max}^2$$

Or $v_{\max} = \sqrt{\frac{2K_{\max}}{m}} = \sqrt{0.1104\times10^{12}}$

$$= 3.323\times10^5\ \text{ms}^{-1} = 332\,\text{kms}^{-1}$$

Q27 **Text Solution:**

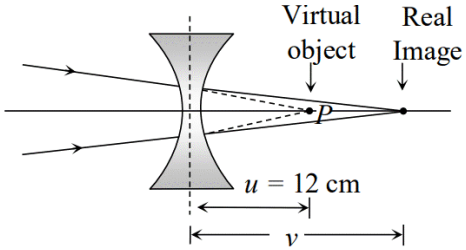
$$\frac{\lambda_1}{\lambda_2} = \frac{\text{n}_2}{\text{n}_1} = \frac{12000}{10000} = \frac{6}{5}$$

$$\text{x} = \frac{\text{n}_1\lambda_1\text{D}}{\text{d}} = \frac{5\times12000\times10^{-10}\times2}{2\times10^{-3}} = 6\text{ mm}$$

Video Solution:

Q28 **Text Solution:**

By using lens formula



$$\frac{1}{-16} = \frac{1}{v} - \frac{1}{(+12)}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{12} - \frac{1}{16} = \frac{4-3}{48}$$

$$\Rightarrow v = 48 \text{ cm}$$

Video Solution:

Q29 **Text Solution:**

According to Rydberg's formula,

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \Rightarrow \lambda \propto \frac{1}{Z^2}$$

$$\frac{1}{3^2} : \frac{1}{2^2} : \frac{1}{1^2} = \frac{1}{9} : \frac{1}{4} : \frac{1}{1} = 4 : 9 : 36$$

Hence, $\lambda_{Li^{2+}} : \lambda_{He^{2+}} : \lambda_H = 4 : 9 : 36$

Video Solution:

Q30 **Text Solution:**

$${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4 + \Delta E$$

The binding energy per nucleon of a deuteron = 1.1 MeV

∴Total binding energy = 2 × 1.1 = 2.2 MeV

The binding energy per nucleon of a helium nuclei = 7 MeV

∴Total binding energy = 4 × 7 = 28 MeV

Hence, the energy released

$\Delta E = (28 - 2 \times 2.2)$ = 23.6 MeV

Video Solution:

Q31 **Text Solution:**

Initial speed $u = 40\, m\, s^{-1}$, angle 30°

Vertical component:

$u_y = u \sin 30^\circ = 20\, m\, s^{-1}$

Speed at ground $v = 1.25u = 50\, m\, s^{-1}$

Using energy/kinematics in the vertical direction:

$$v^2 = u^2 + 2gh$$

$$50^2 = 40^2 + 2(10)h$$

$$2500 - 1600 = 20h \Rightarrow h = 45\, m$$

Q32 **Text Solution:**

A → II B → I C → IV D → III

Q33 **Text Solution:**

(3)

Weight of the bowl = *mg*

$= V\rho g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]\rho g$

Where *D* = Outer diameter *d* = Inner diameter, ρ = Density of bowl Weight of the liquid diplaced by the bowl

$= V\sigma g = \frac{4}{3}\pi\left(\frac{D}{2}\right)^3\sigma g$

where σ is the density of the liquid For the floation

$\frac{4}{3}\pi\left(\frac{D}{2}\right)^3\sigma g = \frac{4}{3}\pi\left[\left(\frac{D}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]pg$

$\Rightarrow \left(\frac{1}{2}\right)^3 \times 1.2 \times 10^3 = \left[\left(\frac{1}{2}\right)^3 - \left(\frac{d}{2}\right)^3\right]2 \times 10^4$

By solving we get *d* = 0.98 m.

Q34 **Text Solution:**

Given:

- $n = 5\,\text{mol}$
- $\Delta T = 120^\circ C - 100^\circ C = 20\,\text{K}$
- $\Delta U = 80\,\text{J}$
- Constant volume → Use:

$$\Delta U = nC_V\Delta T \Rightarrow 80 = 5 \times C_V \times 20$$
$$\Rightarrow C_V = \frac{80}{100} = 0.8\,\text{J/mol\K}$$

Total heat capacity at constant volume (for 5 moles):

$$C_V^{\text{total}} = nC_V = 5 \times 0.8 = 4.0\,\text{J/K}$$

Video Solution:

Q35 **Text Solution:**

$T = 2\pi\sqrt{\frac{I}{C}}$

$I = \frac{2}{5}MR^2$

By equation (i) and (ii)

$T = 2\pi\sqrt{\frac{2MR^2}{5C}}$

Substitute the provided values

$\Rightarrow T = 6.28\sqrt{\frac{0.4\times1\times(0.15)^2}{6\times10^{-3}}}$

$= 6.28\sqrt{\frac{9\times10^{-3}}{6\times10^{-3}}}$

$= 6.28 \times \sqrt{1.5} = 7.7sec.$

Q36 **Text Solution:**

(2)

$U = Bx^4$

$U_{max} = B.A^4$

$k(x) = U_{max} - U(x) = BA^4 - Bx^4$
$= B\left(A^4 - x^4\right)$

$U(x) = \frac{1}{3}K(k)$

$Bx^4 = \frac{1}{3}B\left(A^4 - x^4\right)$

$3Bx^4 = BA^4 - Bx^4$

$4Bx^4 = BA^4 = x^4 = \frac{A^4}{4}$

$x = \pm\frac{A}{\sqrt{2}}$

Video Solution:

Q37 **Text Solution:**

$\frac{d\theta}{dt} \propto \frac{1}{rds} \propto \frac{1}{m^{1/3}}$;

$\frac{R_1}{R_2} = \left(\frac{m_2}{m_1}\right)^{1/3} = \left(\frac{m}{2m}\right)^{1/3} = \frac{1}{2^{1/3}}$

[New NCERT Class 11th Page No. 220]

Video Solution:

Q38 **Text Solution:**

Formula: $V_{rms} = \sqrt{\frac{3RT}{M}}$

V_{rms} = 500 m/s, T = 300 K, R = 8 J/mol-K

$M = \frac{3RT}{V_{rms}^2} = \frac{3\times8\times300}{500^2} = \frac{7200}{250000}$

= 0.0288 kg/mol = 28.8 g/mol

Closest to Nitrogen (N_2, 28 g/mol)

Q39 Text Solution:

Hint:- Area under force displacement is work

$K_f - K_i$ = Work

Solution:-

(1) Incorrect because thfere is acceler^n throughoect so speed will always increase.

(2) $K_f - K_i$ = work

= Area under curve

$= \frac{1}{2}(1)V^2 - \frac{1}{2}(1)(\sqrt{20})^2$

$= \frac{1}{2}(6+2)(10) = 40$

$= \frac{1}{2}(V^2 - 20) = 40$

V = 10 m/s

Correct

(3) Incorrect

(4) Incorrect

Acceleration is constant up to 2m after that it decreases. It is non negative.

Q40 Text Solution:

$a = \frac{6}{3} = 2\,m/s^2$

$\Rightarrow$ velocity of 3 kg after t = 5 sec,

V = (2)(5) = 10 m/s

$\Rightarrow$ velocity of centre of mass

$= \frac{2(0)+3(10)}{2+3} = 6\,m/s$

Video Solution:

Q41 Text Solution:

(4)

AC Generator: It is a device which converts mechanical energy into electrical energy.

Galvanometer: It shows deflection when current passes through it, so it is used to show presence of current in the wire.

Transformer: It is used to step up or stepdown the voltage.

LED: It converts electrical energy to light energy.

(NEW NCERT 12th Page No. 193, 194)

Q42 Text Solution:

(4)

$\frac{\lambda_p}{\lambda_e} = \frac{\frac{h}{m_p v_p}}{\frac{h}{m_e v_e}} = \frac{m_e v_e}{m_p v_p}$

$1.81\times10^{-4} = \frac{9.1\times10^{-31}}{m_p}\times\frac{1}{3}$

$m_p = 1.675\times10^{-27}$ kg

(NEW NCERT 12th Page No. 285)

Video Solution:

Q43 Text Solution:

Field due to a side of length a

$$B_1 = \frac{\mu_0 I}{2\pi\left(\frac{b}{2}\right)}\left(\left(\frac{a}{\sqrt{a^2+b^2}}\right)\right)$$

$$= \frac{\mu_0 I}{\pi}\left(\frac{a}{b\sqrt{a^2+b^2}}\right)$$

Similarly, the field due to a side of length b

$$B_2 = \frac{\mu_0 I}{\pi}\left(\frac{b}{a\sqrt{a^2+b^2}}\right)$$

We can check by using right hand rule that the field due to all the sides will be in the same direction (all perpendicular to the plane of the loop and in the same direction), and hence we can simply add the contributions of the sides

Hence, net field at the centre,

$$B_{NET} = 2B_1 + 2B_2 = \frac{2\mu_0 I}{\pi}\left(\frac{\sqrt{a^2+b^2}}{ab}\right)$$

Video Solution:

Q44 **Text Solution:**

$Y = \overline{\overline{A} + B} = A \cdot \overline{B}$

Q45 **Text Solution:**

All charges are positive and located to the right of the origin, meaning all individual electric fields point in the same direction (negative x-axis). The magnitudes add up to give the net magnitude E_{net}.

$E_{net} = \sum_{n=1}^{10} E_n = \sum_{n=1}^{10} Kn = K\sum_{n=1}^{10} n$

The sum of the first ten integers is calculated as:

$\sum_{n=1}^{10} n = \frac{10(10+1)}{2} = \frac{110}{2} = 55$

Therefore, the net electric field magnitude is

$E_{net} = 55K$ V/m.

N=55

Video Solution:

Q46 **Text Solution:**

Baeyer's reagent (cold alkaline $KMnO_4$) gives syn **1,2-dihydroxylation** of alkenes → vicinal diol from 2-butene.

Video Solution:

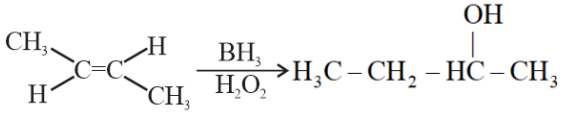
Q47 **Text Solution:**

(III) is most stable as negative charge on oxygen atom.

Video Solution:

Q48 **Text Solution:**

When alkyne reacts with Na/liquid NH_3, trans alkene is formed. (Birch reduction)



Video Solution:

Q49 **Text Solution:**

I and III are **α-alkoxy carbocations (oxocarbenium)** stabilized by resonance from O (I > III by hyperconjugation). II and IV are ordinary alkyl cations (**2° > 1°**). Hence **I > III > II > IV**.

Video Solution:

Q50 **Text Solution:**

$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$

$K_C = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$

Initially	0.0 M	0.35M	0.5 M
at equilibrium	0.0-2x	0.35-x	0.5+2x

$\therefore K_C = \frac{(0.5+2x)^2}{(-2x)^2(0.35-x)}$

$\Rightarrow 4 = \frac{0.25+4x^2+2x}{4x^2(0.35-x)}$

$\Rightarrow 16x^2(0.35-x) = 0.25+4x^2+2x$

$\Rightarrow 5.6x^2 - 16x^3 = 0.25+4x^2+2x$

$\Rightarrow 16x^3 - 1.6x^2 + 2x + 0.25 = 0$

$\Rightarrow x = -0.10633$

∴Concentration of O_2 at equilibrium

$= 0.35 - (-0.10633)$

= 0.35 + 0.10633

= 0.456 M

Video Solution:

Q51 **Text Solution:**

At A, temperature $= 300$ K, volume $= 10$ L,

Pressure $= P_1$

At C, Temperature $= 600$ K, volume $= 20$ L,

Pressure $= P_2$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow \frac{P_1 \times 10}{300} = \frac{P_2 \times 20}{600}$$

$P_1 = P_2$

Thus; the process is isobaric.

Video Solution:

Q52 **Text Solution:**

$-CH_3$, $-CH_2CH_3$ groups are e^- donating.

Video Solution:

Q53 **Text Solution:**

$[Ag^+] = s$

$[Cl^-] = s + 0.1 \approx 0.1$

$K_{sp} = [Ag^+][Cl^-]$

$1.2 \times 10^{-10} = s \times 0.1$

$\Rightarrow s = 1.2 \times 10^{-9}$

Video Solution:

Q54 **Text Solution:**

(1)

Video Solution:

Q55 **Text Solution:**

$\Delta x.\ m\Delta v = \frac{h}{4\pi}$

$\Delta x.\ \Delta v = \frac{h}{4\pi m} = \frac{6.63\times10^{-34}}{4\times 3.14\times9.1\times10^{-31}} = 5.8 \times 10^{-5}\ m^2/s$

Video Solution:

Q56 **Text Solution:**

30

Video Solution:



Q57 Text Solution:

Meso compound possess either plane or centre of symmetry. They do not possess both plane of symmetry and centre of symmetry and also they are internally compensated form.

[New NCERT Class 11th Page No. 270]

Video Solution:



Q58 Text Solution:

(3)

Video Solution:



Q59 Text Solution:

(3)

Video Solution:



Q60 Text Solution:

Step 1: Count total electrons

1. NO_3^-

- N = 5, O = 6×3 = 18, +1 extra electron (negative charge) = 24 electrons

2. CO_3^{2-}

- C = 4, O = 6×3 = 18, +2 extra electrons = 24 electrons

3. ClO_3^-

- Cl = 17, O = 6×3 = 18, +1 extra electron = 36 electrons → not isoelectronic with NO_3^- or CO_3^{2-}

4. SO_3

- S = 16, O = 6×3 = 18 → 34 electrons → not matching

✅ So NO_3^- and CO_3^{2-} are isoelectronic.

Step 2: Check structures

- Both NO_3^- and CO_3^{2-}: trigonal planar → isostructural

✅ Correct answer:

Option (1) — NO_3^-, CO_3^{2-}

Video Solution:



Q61 Text Solution:

(4)

Q62 Text Solution:

The expansion described is expansion into vacuum, which is called free expansion.

Work done in expansion is given by

$w = -P_{ext} \Delta V$

where

P_{ext} = external pressure.

In free expansion

$P_{ext} = 0$

Therefore

$w = -0 \times \Delta V$

$w = 0$

Now for an ideal gas in isothermal process

$\Delta U = 0$

From the first law of thermodynamics

$\Delta U = q + w$

$0 = q + 0$

Therefore

$q = 0$

Thus no heat is absorbed and no work is done.

Q63 Text Solution:

1

Q64 Text Solution:

$n = 4$ $l = 1$

$l = 1$ shows the p-subshell.

The maximum number of electrons will be six.

Video Solution:

Q65 Text Solution:

$$\begin{array}{lccc} & N_2 & + \quad 3H_2 & \rightarrow \quad 2\,NH_3 \\ \text{Initial} & 2 \text{ mole} & 6 \text{ mole} & 0 \\ \text{Eq} & \left(2-\frac{50}{100}X2\right) & (6-3x1) & 2x1 \\ & 1 & 3 & 2 \end{array}$$

$$k_c = \frac{[NH_3]^2}{[N_2]\,[H_2]^3} = \frac{\left(\frac{2}{1}\right)2}{\left[\frac{1}{1}\right]\left[\frac{3}{1}\right]^3} = \frac{4}{27}$$

Video Solution:

Q66 Text Solution:

By writing the equation for the reverse reaction

$N_2O_5\,(g) \rightarrow 2\,NO_2\,(g) + \frac{1}{2}O_2\,(g)$

New pressure constant Kp for this reaction

$$K_p = \frac{1}{K_p}$$

$$K_p = \frac{[NO_2]^2[O_2]^{\frac{1}{2}}}{[N_2O_5]}$$

Thus pressure constant for the reaction,

$$2N_2O_5\,(g) \rightleftharpoons 4\,NO_2\,(g) + O_2\,(g)$$

$$K_p'' = \frac{[NO_2]^4[O_2]}{[N_2O_5]^2}$$

(New NCERT Class 11th Page No. 182)

Video Solution:

Q67 Text Solution:

Molarity of H_2SO_4 = 18 M

Moles of H_2SO_4 = 18

Volume of H_2SO_4 = 1000 mL

Mass of H_2SO_4 = 18 × 98 = 1764 g

Mass of solution = 1.8 × 1000 = 1800 g

Mass of solvent = 1800 – 1764 = 36 g

$$\text{Molality} = \frac{\text{moles of } H_2SO_4}{\text{mass of solvent}} \times 1000$$

$$\text{Molality} = \frac{18}{36} \times 1000 = 500 \text{ m}$$

Video Solution:

Q68 Text Solution:

(2)

Video Solution:

Q69 Text Solution:

The given reaction is:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2$

1 mole of Zn reacts with 2 mol of HCl

to produce 1 mol of H_2,0.3 mol of Zn will

react with 0.6 mol of HCl but there is only

0.52 mol of HCl. So, HCl is the limiting reagent.

2 mol of HCl gives 1 mol of H_2

So, 0.52 mol of HCl will give 0.26 mol of H_2

Video Solution:

Q70 **Text Solution:**

(4)

Video Solution:

Q71 **Text Solution:**

(4)

$\Delta S = -ve$

$\Delta H = -ve$

$\Delta G = \Delta H - T\Delta S$

For a reaction to be spontaneous; $\Delta G = -ve$ which can be below a certain temperature only

[New NCERT Class 11th Page No. 160]

Video Solution:

Q72 **Text Solution:**

$Zn \rightarrow Zn^{2+} + 2e^-$

$E = E^{\circ}_{O.P.} - \frac{0.0591}{n}\log\left[Zn^{2+}\right]$

$= 0.763 - \frac{0.0591}{2}\log\ 0.01$

$= 0.763 + 0.0591 = 0.8221\ V$

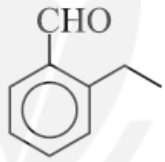
Video Solution:

Q73 **Text Solution:**

Boiling point falls as branching increases: the unbranched primary alcohol (1-pentanol) has the greatest surface area and highest bp, followed by the straight-chain secondary (2-pentanol). The branched secondary (3-methyl-2-butanol) boils lower, and the highly branched tertiary alcohol (2-methyl-2-butanol) is lowest.

Video Solution:

Q74 **Text Solution:**



$\Rightarrow$ + 2, 4 DNP Test
+ Tollen's Test
+ Cannizzaro reaction

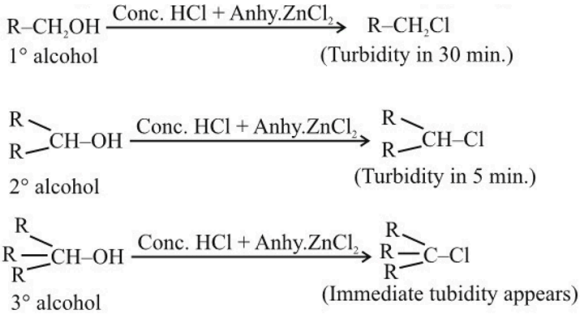
(New NCERT 11th Page No. 154)

Video Solution:

Q75 **Text Solution:**

(1)

Yes, Lucas Reagent is a mixture of anhydrous $ZnCl_2$ (Zinc Chloride) and HCl(Hydrochloric Acid). Hydrochloric Acid is used for the distinction of monohydric alcohol. Tertiary alcohols on addition produce a precipitate immediately while secondary alcohols produce ppt. after 5 minutes. Primary alcohols do not produce any precipitate.

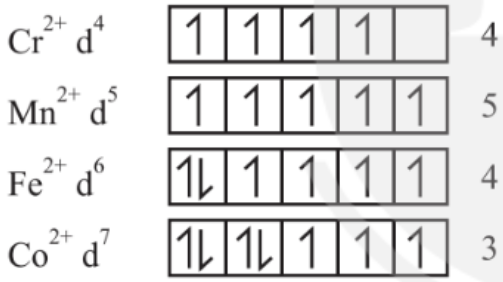


Therefore, assertion is true but reason is false.

Video Solution:

Q76 **Text Solution:**

As magnetic moment is $\sqrt{n(n+2)}$ where n is no. of unpaired electron. So ion with minimum no. of unpaired electron will have minimum paramagnetism.



Video Solution:

Q77 **Text Solution:**

A S_N2 reaction proceeds with complete stereochemical inversion while a S_N1 reaction proceeds with racemization.

Video Solution:

Q78 **Text Solution:**

Benzenesulphonyl chloride $\rightarrow$ Hinsberg reagent

Hoffmann bromamide reaction $\rightarrow$ Known reaction of Isocyanates

Carbylamine reaction $\rightarrow$ Test for primary amines

Hoffmann orientation $\rightarrow$ Anti Saytzeff

Video Solution:

Q79 **Text Solution:**

$$Ag^{+}(aq) + 2S_2O_3^{2-}(aq) \rightarrow \underbrace{[Ag(S_2O_3)_2]^{3-}(aq)}_{\text{स्पीशीज X (साफ विलयन)}}$$

$$2\left[Ag(S_2O_3)_2\right]^{3-}(aq) + 3Ag^{+}(aq) \rightarrow 2\underbrace{Ag_2S_2O_3(s)}_{\text{स्पीशीज Y (श्वेत अवक्षेप)}}\downarrow + S_2O_3^{2-}(aq)$$

$$Ag_2S_2O_3(s) + H_2O(l) \xrightarrow{\text{"समय के साथ)}} \underbrace{Ag_2S(s)}_{\text{"स्पीशीज Z (काला अवक्षेप)}}\downarrow + H_2SO_4(aq)$$

Video Solution:

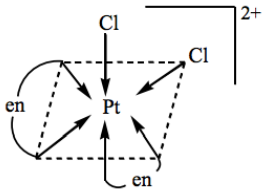
Q80 **Text Solution:**

$2AgCl \xrightarrow{hv} 2Ag \downarrow (black) + Cl_2 \uparrow$.

Video Solution:

Q81 **Text Solution:**

$cis - [PtCl_2(en)_2]^{2+}$ is optically active, due to absence of plane of symmetry.



Video Solution:

Q82 **Text Solution:**

(4)

Video Solution:

Q83 **Text Solution:**

- **Statement I is correct:** Cell potential E_{cell} is an intensive property—it does not depend on the quantity of material, only on temperature, pressure and composition.
- **Statement II is incorrect:** Resistivity (ρ) has SI units Ω m, whereas Ω^{-1} m (or S m^{-1}) is the unit of conductivity.

Video Solution:

Q84 **Text Solution:**

The central metal ion is Cobalt in the +3 oxidation state. There are four ammonia molecules as ligands, which are denoted by ammine. There is one water molecule as a ligand, denoted by aqua.

The complex is an ionic compound, so the counter-ion is chloride, and the formula suggests there is one chloride ion for every cationic complex.

[NCERT 12th Edition 2024-25 Page No. 129]

Video Solution:

Q85 **Text Solution:**

$$Cr(OH)_3 + OH^- \rightleftharpoons \left[Cr(OH)_4\right]^-$$

$$2\left[Cr(OH)_4\right]^- + 2OH^- + 3H_2O_2$$

$$\longrightarrow 2CrO_4^{2-}\,(\text{yellow coloured solution})$$

$$+ 8H_2O$$

$$Fe(OH)_3 + NaOH/H_2O_2 \longrightarrow \text{no change}$$

Video Solution:

Q86 **Text Solution:**

(3)

$$CH_2 = CH_2 + O_2 \xrightarrow[H_2O]{Pd(II)/Cu(II)} CH_3-\overset{\overset{O}{\|}}{C}-H$$

(Acetaldehyde)

$$CH_3 - CH_2 + OH \xrightarrow{Cu/573k} H_3C-\overset{\overset{O}{\|}}{C}-H$$

(Acetaldehyde)

$$CH_3\,CH_2\,OH \xrightarrow{CrO_3-H_2\,SO_4} CH_3COOH$$

$$CH_3 - CN \xrightarrow[(2)\,H_2O]{(1)\,DiBAL-H} CH_3-\overset{\overset{O}{\|}}{C}-H$$

(Acetaldehyde)

Video Solution:

Q87 **Text Solution:**

For reaction, $2A \rightarrow 3B$

$$-\frac{1}{2}\frac{d[A]}{dt} = \frac{1}{3}\frac{d[B]}{dt}$$

$$\frac{d[B]}{dt} = \times \text{mol}/L$$

$$\frac{d[A]}{dt} = -\frac{2}{3} \Rightarrow \frac{d[B]}{dt} = -\frac{2}{3}x$$

Hence, the rate of consumption of

$$A = \left[\frac{-d[A]}{dt} = \frac{2}{3}x\right]$$

Video Solution:

Q88 **Text Solution:**

Correct option is (2).

- A. $[Co(NH_3)_6]^{3+}$ → Yellowish orange (4)
- B. $[Ti(H_2O)_6]^{3+}$ → Violet (3)
- C. $[Ni(H_2O)_6]^{2+}$ → Green (2)
- D. $[Ni(H_2O)_4(en)]^{2+}$ → Blue (1)

Video Solution:

Q89 **Text Solution:**

$$\Lambda_m^C = 7.814 \times 10^{-4}\ \Omega^{-1}m^2\ mol^{-1}$$

$$\Lambda_m^\circ = 3.907 \times 10^{-2}\ \Omega^{-1}m^2\ mol^{-1}$$

$$\alpha = \frac{\Lambda_m^C}{\Lambda_m^\circ} = \frac{7.814\times10^{-4}}{3.907\times10^{-2}} = 2 \times 10^{-2}$$

So, $K_a = \frac{C\alpha^2}{1-\alpha}$

Since, $\alpha << 1$, so neglect α from denomin

$$K_a = C\alpha^2 = 0.05 \times \left(2 \times 10^{-2}\right)^2$$

$$K_a = 0.05 \times 4 \times 10^{-4} = 20 \times 10^{-6}$$

$$K_a = 2 \times 10^{-5}$$

Video Solution:

Q90 **Text Solution:**

Phosphorus forms $d\pi - d\pi$ bonds with transition metals and has a hydride (PH_3) that is a stronger base compared to other group 15 hydrides that form $d\pi - d\pi$ bond like AsH_3, SbH_3.

Video Solution:

Q91 Text Solution:

Statement (A) is CORRECT: Bacteria, *Mycoplasma*, and cyanobacteria (blue-green algae) are all unicellular organisms lacking a membrane-bound nucleus and organelles, making them prokaryotes (Kingdom Monera).
Statement (B) is CORRECT: Slime moulds (*Physarum* is a genus of slime moulds) and Euglenoids belong to Kingdom Protista, which consists of unicellular eukaryotes.
Statement (C) is INCORRECT: Mushrooms (*Agaricus*), toadstools, rusts (*Puccinia*), and smuts (*Ustilago*) belong to Basidiomycetes (club fungi), not Ascomycetes (sac fungi).
Statement (D) is INCORRECT: While viruses and viroids are non-cellular/acellular infectious entities, lichens are symbiotic organisms composed of a fungus (mycobiont) and an alga/cyanobacterium (phycobiont), both of which are cellular.
Thus, statements (A) and (B) are correct.
Hence, the correct answer is Option A.

Q92 Text Solution:

(1)

Q93 Text Solution:

(2)

Q94 Text Solution:

(4)

Q95 Text Solution:

Enzymes involved in DNA replication:

- DNA polymerase
- DNA ligase
- Helicase
- Topoisomerase (DNA gyrase in prokaryotes)
- Primase

The remaining proteins in the question (**β-galactosidase** and **Permease**) belong to the **lac operon**, while **Peptidyl transferase** is involved in **protein synthesis (translation)**.

Q96 Text Solution:

(4)

Q97 Text Solution:

(4)

Q98 Text Solution:

(3)

Q99 Text Solution:

Harmful mutations often remain in the gene pool because they are recessive and do not manifest in heterozygous individuals. These carriers can pass the mutation to offspring without showing any symptoms, allowing the mutation to persist in the population.

Q100 Text Solution:

(2)

Video Solution:

Q101 Text Solution:

(1)

Video Solution:

Q102 Text Solution:

Two moleules of ATP are synthesised during glycolysis and total 38 ATP are synthesised by complete oxidation of glucose molecule in eukaryotes.

Q103 Text Solution:

(2)

Video Solution:

Q104 Text Solution:

Explanation: Succinate dehydrogenase is part of Complex II in ETS and directly transfers electrons from $FADH_2$ to ubiquinone (CoQ).

Q105 Text Solution:

(1)

The Abingdon tortoise in Galapagos Islands became extinct within a decade after goats were introduced on the island, apparently due to the greater browsing efficiency of the goats. This is an example of competition.

Q106 Text Solution:

(1)

Interspecific interactions arise from the interaction of populations of two different species. They could be beneficial, detrimental or neutral (neither harm nor benefit) to one of the species or both. Commensalism is the interaction in which one species benefits and the other is neither harmed nor benefited.

(NEW NCERT 12th Page No. 196, 201)

Video Solution:

Q107 Text Solution:

Carrying capacity is the maximum population an environment can sustain; growth slows down due to limited resources and competition.

Q108 Text Solution:

Fungi prefer to grow in warm and humid places.

[New NCERT Class 11th Page No. 17]

Q109 Text Solution:

(1)

A virus is a nucleoprotein and the genetic material is infectious.

[New NCERT Class 11th Page No. 19]

Q110 Text Solution:

The ethical argument for conserving biodiversity relates to what we owe to millions of plant, animal and microbe species with whom we share this planet.

Philosophically or spiritually, we need to realise that every species has an intrinsic value, even if it may not be of current or any economic value to us.

Video Solution:

Q111 Text Solution:

Refer to video solution

Video Solution:

Q112 Text Solution:

1. Susceptibility to extinction is not always the same for all species.

2. It is related to various life history characteristics that influence the species' vulnerability to human activities and natural

surroundings.
3. Species that occur in a single geographic area are more vulnerable to extinction.
4. This is because a loss of habitat in one geographic location will result in a complete loss for the species.
5. The species that are more susceptible to extinction include the following characteristics: Large body size, small population, low reproductive rate, narrow range of distribution, poor dispersal rate etc.
6. Hence the correct option then is species restricted to a small geographical area.

Video Solution:



Q113 Text Solution:

Bivalents become clearly visible in Pachytene, not Zygotene. In Zygotene, homologous chromosomes begin pairing and synapsis occurs.

Video Solution:



Q114 Text Solution:

A bivalent, or tetrad, consists of a pair of homologous chromosomes, each made up of two sister chromatids, totaling four chromatids and two centromeres.

Video Solution:



Q115 Text Solution:

(3)

In dicotyledonous plants and gymnosperms, the lateral meristems, vascular cambium, and cork-cambium appear later in life. These are the meristems that cause the increase in the girth of the organs in which they are active. This is known as secondary growth of the plant.

[New NCERT Class 11th Page No. 167-168]

Q116 Text Solution:

(2)

Absolute growth rate is defined as the measurement and the comparison of total growth per unit time.

[New NCERT Class 11th Page No. 171]

Q117 Text Solution:

In majority of higher animals and plants, growth and reproduction are mutually exclusive events. One must remember that increase in body mass is considered as growth. Non-living objects also grow if we take increase in body mass as a criterion for growth. Mountains, boulders and sand mounds do grow. However, this kind of growth exhibited by non-living objects is by accumulation of material on the surface. In living organisms, growth is from inside. Growth, therefore, cannot be taken as a defining property of living organisms. Reproduction, likewise, is a characteristic of living organisms. Further, there are many organisms which do not reproduce (mules, sterile worker bees, infertile human couples, etc). Hence, reproduction also cannot be an all-inclusive defining characteristic of living organisms. While metabolism is a defining feature of all living organisms without exception, isolated metabolic reactions in vitro are not living things but surely living reactions. Human being is the only organism who is aware of himself, i.e., has self-consciousness.

Q118 Text Solution:

The scientific name of mango is written as *Mangifera indica.*

Q119 **Text Solution:**

(2)

In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. No organelles, like the ones in eukaryotes, are found in prokaryotic cells except for ribosome. A special membranous structure in bacteria is the mesosome which is formed by the extensions of plasma membrane into the cell. The eukaryotes include all the protists, plants, animals and fungi.

[NCERT 11th Edition 2025-26 Page No. 88, 90, 91]

Q120 **Text Solution:**

Plasmodesmata are cytoplasmic bridges between neighboring cells, showing pores between the primary cell wall and middle lamella.

Video Solution:

Q121 **Text Solution:**

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(3)

2. Neutral solutes may move across the membrane by the process of simple diffusion along the concentration gradient.

3. The quasi-fluid nature of lipid enables lateral movement of proteins within the overall bilayer.

Q122 **Text Solution:**

Leaching is the process by which water-soluble inorganic nutrients **move downward** in the soil horizon, not upward. These nutrients percolate through the soil and may get precipitated as unavailable salts.

Video Solution:

Q123 **Text Solution:**

(3)

Video Solution:

Q124 **Text Solution:**

Stratification in an ecosystem refers to the **vertical distribution of different species occupying different levels**, which makes **statement A correct**. In a forest ecosystem, this vertical layering is clearly visible: **trees form the topmost vertical strata**, making **statement B correct**. **Grasses occupy the bottom layer** close to the ground, so **statement D is also correct**. However, **statement C is incorrect** because herbs do **not always occupy the second layer**. Typically, **shrubs form the second layer**, while **herbs and grasses occupy the lower layers**. Since statement C gives an inaccurate placement, it cannot be included as correct.

Therefore, the correct set of statements is **A, B, and D**, making **Option A** the correct choice.

Video Solution:

Q125 **Text Solution:**

(1)

Bottled fruit juices bought from the market are clearer as compared to those made at home.

Bottled juices are clarified by the use of proteases and pectinases.

(NEW NCERT 12th Page No. 153)

Q126 Text Solution:

(4)

Video Solution:

Q127 Text Solution:

Monascus purpureus is prepared commercially as a blood cholesterol-lowering medication.

- It is a red yeast.
- It produces Statins.
- Reddish-purple *Monascus purpurius* yeast is a kind of yeast.
  Statin is produced by this yeast.
- This is used in industries on a commercial basis as a result.
- Drugs like statin are used to decrease cholesterol levels in the blood.
- The HMG-Co-A reductase enzyme is inhibited by this.
- Lipid synthesis is carried out by HMG-Co-A reductase. The enzyme's competitive inhibitor is statin.

Q128 Text Solution:

Classes	Common Name	Major Pigments	Stored Food
Chlorophyceae	Green algae	Chlorophyll *a, b*	Starch
Phaeophyceae	Brown algae	Chlorophyll *a, c,* fucoxanthin	Mannitol, laminarin
Rhodophyceae	Red algae	Chlorophyll *a, d,* Phycoerythrin	Floridean starch

[New NCERT Class 11th Page No. 27]

Q129 Text Solution:

(3)

Video Solution:

Q130 Text Solution:

(4)

Video Solution:

Q131 Text Solution:

(1)

Video Solution:

Q132 Text Solution:

- Metaphloem lacks phloem fibres.
- It contains sieve tubes.
- It is not formed from vascular cambium as it is a part of primary phloem.

Video Solution:

Q133 Text Solution:

- Monocot plants cannot be grafted because they lack the cambium tissue found in dicot plants.

- Cambium tissue is a meristematic tissue present in the vascular bundles of dicot plants.
- This tissue has a propensity for regeneration, which means it can grow new tissue during grafting.
- Because monocots lack cambium, grafting is difficult.
- Only dicots and gymnosperms, not monocots, are affected.
- Monocots are unable to graft due to a lack of cambium, not because of scattered vascular bundles, parallel venation, or herbaceousness.

Q134 **Text Solution:**
(2)

Video Solution:

Q135 **Text Solution:**
C_4 plants are use to grow in dry tropical provinces. Sugarcane, Maize etc., are examples of C_4 plants. In C_4 plants the Calvin cycle takes place in the bundle-sheath cells. C_4 plants have a special type of structural organization in their leaf known as Kranz anatomy. They can tolerate higher temperatures on presence of such kind of structures.

Video Solution:

Q136 **Text Solution:**
(3)

Video Solution:

Q137 **Text Solution:**
PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption. Nearly all of the essential nutrients, and 70-80 per cent of electrolytes and water are reabsorbed by this segment. PCT also helps to maintain the pH and ionic balance of the body fluids by selective secretion of hydrogen ions, ammonia and potassium ions into the filtrate and by absorption of HCO_3^- from it.

Q138 **Text Solution:**
(4)

Video Solution:

Q139 **Text Solution:**

List-I (Organisms)	List-II (Respiratory organisms)
Shark	Gills
Insects	A network of tracheal tubes
Bat	Lungs
Earthworm	Moist skin

Video Solution:

Q140 **Text Solution:**

3

Video Solution:

Q141 **Text Solution:**

Option (2) is the correct answer. NaCl is transported by the ascending limb of Henle's loop and exchanged with the descending limb of vasa recta. This process helps in maintaining the concentration gradient in the renal medulla. The ascending portion of vasa recta returns NaCl to the interstitium. On the other hand, small amounts of urea enter the thin segment of the ascending limb of Henle's loop and are transported back to the interstitium by the collecting tubule. This intricate process ensures efficient reabsorption of NaCl and urea in the renal system, contributing to the regulation of water and electrolyte balance in the body. Understanding the transport mechanisms in the renal system is crucial for comprehending the overall functioning of the kidneys.

Video Solution:

Q142 **Text Solution:**

About 97 per cent of O_2 is transported by RBCs in the blood. The remaining 3 per cent of O_2 is carried in a dissolved state through the plasma. This percentage of oxygen in plasma is comparable to the proportion of eosinophils in total WBCs (typically 2-3%)

Q143 **Text Solution:**

- **Thrombokinase** (also known as **thrombokinase or prothrombinase**) is an enzyme that plays a crucial role in **blood coagulation (clotting)**.
- It catalyzes the conversion of **prothrombin into thrombin**, which then helps in the formation of a **fibrin clot**.
- This process is essential for **stopping bleeding and initiating wound healing**.

Why other options are incorrect:

- **A. Production of erythrocytes from the bone marrow** → This is the process of **erythropoiesis**, regulated by the hormone **erythropoietin**, not thrombokinase.
- **B. Pulmonary and systemic circulation** → This involves the movement of blood between the heart, lungs, and body but is unrelated to thrombokinase.
- **C. Cardiac cycle and its regulation** → The cardiac cycle is controlled by electrical impulses (SA node, AV node) and hormonal regulation, not thrombokinase.

Video Solution:

Q144 **Text Solution:**

1

Video Solution:

Q145 Text Solution:

(2)

Video Solution:

Q146 Text Solution:

(4)

6-15 pairs of gill slits	*Myxine*
Gill slits are without operculum.	Chondrichthyes
Ctenoid scales	Osteichthyes
Pneumatic bones	*Corvus*

[New NCERT Class 11th Page no. - 47, 48, 49, 50]

Video Solution:

Q147 Text Solution:

During dehydration, there is a loss of body fluid volume and an increase in ionic concentration in the blood. This activates osmoreceptors in the hypothalamus, which stimulate the release of ADH from the posterior pituitary. ADH promotes water reabsorption in the kidneys, reducing urine output and conserving water. ADH also causes vasoconstriction, which helps maintain blood pressure and glomerular filtration rate (GFR).

(NEW NCERT 11th Page No. 212)

Q148 Text Solution:

1). The process by which the nephron takes water and solutes from the tubular fluid (pre-urine) and returns them to the circulating blood is known as reabsorption or tubular reabsorption in the field of renal physiology.

2). Reabsorption is a crucial process since the filtrate includes a lot of beneficial compounds that must be sent to the blood in order for the body to utilise them. The beneficial substances would exit the body through the urine if there is no tubular reabsorption in the nephrons (urea and water formed urine).

3). Selective reabsorption enables beneficial substances to enter capillaries once again where the body will employ them for subsequent processes. Poor renal reabsorption will result in excessive fluid loss and a lack of fluid inside the body. Therefore, a mechanism for fluid conservation should be used.

4). Anti-diuretic hormone works on the kidneys and blood arteries to lower blood pressure. Its primary function is to reduce the amount of water excreted in the urine, so conserving the volume of fluid in your body. Thus, its growth will be beneficial.

5). Reduced glomerular filtration will result in less urine being produced, saving the body's water supply. Reduced arterial pressure will result in decreased blood flow to the kidney, resulting in low GFR and a reduction in urine production. These three measures will therefore conserve water.

6). This patient won't benefit from higher arterial pressure because it may increase renal blood flow and GFR, which would result in excessive volume loss from the body.

Video Solution:

Q149 Text Solution:

The correct answer is:
(2) Unbranched chain of glucose molecules linked by β-1,4 glycosidic bonds
Explanation:

- **Cellulose** is a **structural polysaccharide** composed of **β-D-glucose units** linked by **β-1,4-glycosidic bonds**.
- It forms **long, unbranched chains** that align to create **strong hydrogen bonding networks**, making cellulose **rigid and insoluble**.
- It is the **main component of plant cell walls**.

Why other options are incorrect:

- **(A) α-1,6 glycosidic bonds** → Found in **glycogen and amylopectin**, not cellulose.
- **(C) Branched chain with α-1,6 bonds** → Incorrect because **cellulose is unbranched**.
- **(D) Combination of α-1,6 and β-1,4 bonds** → Incorrect because **cellulose only has β-1,4 bonds**.

Thus, **cellulose is an unbranched polymer of glucose linked by β-1,4-glycosidic bonds, making option (2) correct**.

Video Solution:

Q150 **Text Solution:**
(4)

- Enzymes isolated from thermophilic organisms are stable and retain their catalytic power even at high temperatures (upto 80°-90°C).
- The active site of an enzyme breaks the chemical bonds of the substrate, not the product.

[New NCERT Class 11th Page No. 113, 115]

Q151 **Text Solution:**

Cockroach	Malpighian tubules
Earthworm	Nephridia
Planaria	Flame cells
Prawn	Antennal glands

[New NCERT Class 11th Page No. 206]

Video Solution:

Q152 **Text Solution:**
(4)

- Human circulatory system, also called the blood vascular system consists of a muscular chambered heart, a network of closed branching blood vessels and blood, the fluid which is circulated.
- Heart is derived from mesoderm.
- The walls of ventricles are much thicker than that of the atria.
- In birds and mammals, oxygenated and deoxygenated blood received by the left and right atria respectively passes on to the ventricles of the same sides.

[New NCERT Class 11th Page No. 197,198, 199]

Video Solution:

Q153 **Text Solution:**

- **Starch** is **chemically nonreactive** (B), preventing degradation.
- It is **osmotically inactive** (D), preventing water imbalance in cells.

Video Solution:



Q154 **Text Solution:**

(3)

Video Solution:



Q155 **Text Solution:**

- Adrenal medullary hormones can increase cardiac output.
- RBCs have an average life span of 120 days.

Video Solution:



Q156 **Text Solution:**

(1)

Aldosterone acts mainly at distal part of the renal tubule and stimulates the reabsorption of Na^+ and water and excretion of K^+ and phosphate ion.

[New NCERT class 11 page number - 232]

Video Solution:



Q157 **Text Solution:**

(2)

Video Solution:



Q158 **Text Solution:**

(4)

Video Solution:



Q159 **Text Solution:**

***Plasmodium*, a small protozoan**, is the cause of malaria.

Plasmodium enters the human body as **sporozoites**, which are spread by **female *Anopheles* bites.**

- To complete its life cycle, the malaria parasite needs **two hosts: > Humans and > Animals.**

Mosquito (female *Anopheles*): vector or agent of transmission

***Plasmodium falciparum* is responsible for cerebral malignant malaria.**

Plasmodium ovale, which is found in West Africa and South America, is one of several plasmodium species that attack people.

- The incubation period is **14 days**, and it produces moderate tertian malaria.

Plasmodium vivax produces benign tertian malaria by causing schizonts and gametocytes to expand infected erythrocytes.

- As a result of the production of new merozoites, the fever occurs every **48 hours**.
- It has a 14-day incubation period and thrives in tropical and temperate climates.

Plasmodium malariae: This species causes quartan malaria. Schizonts and gametocytes make erythrocytes very large.

- It has a **27-to-30-day** incubation period.
- Every **72 hours**, the fever returns, and there are relapses.

Plasmodium falciparum is the most lethal of all plasmodium species. Erythrocytes are not enlarged.

- It is found in the tropics and produces **cerebral malignant malaria or sub-tertian malaria fever, both of which are lethal.**
- There is a **12-day incubation** period.
- RBCs cluster together to form masses, obstructing the brain's blood vessels.
- *Plasmodium falciparum* can also induce black water fever (pernicious malaria), which causes RBCs to be destroyed in large numbers.

As a result, **option 3 is the right response**.

Q160 Text Solution:

1. A woman's body goes through a monthly series of changes in order to get prepared to be able to give birth to a child.
2. By each month, the ovulation takes place simultaneously with the hormonal changes which prepare the uterus for childbearing.
3. When ovulation happens and the egg is not fertilized, that is, when the gametes have not fused, the lining of the uterus is caused to be shed through the vagina end.
4. The menstrual cycle can be divided into different phases based on the events as occurred in the ovary (ovarian cycle) and the uterus (uterine cycle).
5. During the Proliferative phase, the primary follicles in the ovary grow, and simultaneously the endometrium of the uterus regenerates.
6. In the proliferative phase, the endometrium of the uterus becomes thicker as blood vessels become long and coiled.
7. These changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones.
8. Estrogen creates a new layer of the endometrium (the inner epithelial layer of the uterus) in the uterus which is denoted as the proliferative endometrium.

So, the correct option is (3) i.e.Blood vessel in the endometrium becomes long and coiled

Video Solution:



Q161 Text Solution:

(2)

Implants are used under skin.

Multiload 375 is a copper releasing IUD and implants have six matchstick sized tubes containing hormones.

Mode of action of implants is similar to that of pills.

Effective period of implants is shorter than pills.

Emergency contraceptives could be used to avoid possible pregnancy due to rape.

Oral Contraceptive pills inhibit ovulation.

(New NCERT 12th Page no.-45)

Q162 Text Solution:

(1)

Video Solution:



Q163 Text Solution:

- Statement I is **true**. Each ovary is covered by a thin germinal epithelium, which encloses the ovarian stroma.
- Statement II is **false**. The ovarian stroma is divided into an **outer cortex** and an **inner medulla**, not the other way around.

- Statement III is **false**. Cyclical changes during the menstrual cycle occur in the **endometrium**, not in the myometrium. The myometrium is the muscular layer of the uterus and helps in uterine contractions.
- Statement IV is **true**. The male reproductive system is located in the **pelvic region** and includes testes, ducts, accessory glands, and external genitalia.

Thus, the correct sequence is: I–True, II–False, III–False, IV–True.

Video Solution:

Q164 **Text Solution:**

(3)

Video Solution:

Q165 **Text Solution:**

(2)

- The sex of a child is determined by the combination of sex chromosomes inherited from the parents.
- A female has two X chromosomes (XX), and all her eggs carry an X chromosome.
- A male has one X and one Y chromosome (XY), and his sperm can carry either an X or a Y chromosome.
- If an X-carrying sperm fertilises the egg, the child will be female (XX).
- If a Y-carrying sperm fertilises the egg, the child will be male (XY).
- Therefore, it is the type of sperm from the male parent that determines the sex of the offspring.

Video Solution:

Q166 **Text Solution:**

(3)

- The Green Revolution succeeded in tripling the food supply but yet it was not enough to feed the growing human population.
- Increased yields have partly been due to the use of improved crop varieties, but mainly due to the use of better management practices and use of agrochemicals (fertilisers and pesticides).
- However, for farmers in the developing world, agrochemicals are often too expensive, and further increases in yield with existing varieties are not possible using conventional breeding.

[New NCERT Class 12th Page No. 178]

Video Solution:

Q167 **Text Solution:**

After the transformation of spermatids into sperm, their heads become embedded in a cell called **sertoli cells** (X) and are finally released from the **seminiferous tubule** (Y) by the process called **spermiation** (Z).

Hence, the correct option is '3'.

Q168 **Text Solution:**

(2)

Video Solution:

Q169 **Text Solution:**

(4)

Video Solution:

Q170 **Text Solution:**

(2)

Video Solution:

Q171 **Text Solution:**

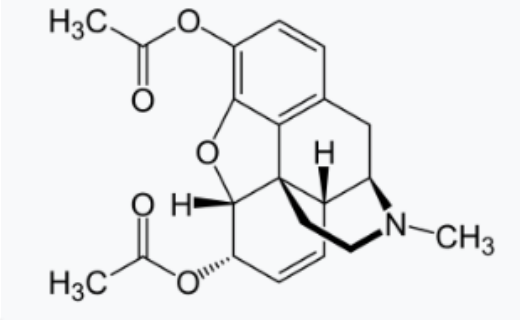
(2)

Video Solution:

Q172 **Text Solution:**

1. Opioids are drugs that bind to specific opioid receptors present in our central nervous system and gastrointestinal tract.
2. Heroin, which is chemically called diacetylmorphine is also known as smack.
3. It is a bitter and crystalline compound.
4. It is white in color and odorless.
5. It is a semi synthetic compound obtained from the acetylation of morphine that is extracted from the latex of the poppy plant.
6. It is generally taken by snorting or injection.

Heroin is a depressant and slows down body functions.



Hence the answer is (4) heroin.

Q173 **Text Solution:**

During intense lactation, the hormone prolactin is secreted in large amounts to stimulate milk production. High levels of prolactin inhibit the release of gonadotropins (LH and FSH) from the pituitary gland. Since gonadotropins are essential for ovulation, their suppression prevents the release of an egg, thereby acting as a natural contraceptive for a few months after childbirth.

Video Solution:

Q174 **Text Solution:**

(1)

Video Solution:

Q175 **Text Solution:**

(4)

Video Solution:



Q176 **Text Solution:**

Codeine partly gets converted into morphine ,both belongs to similar class of drugs called opiods having analgesic and sedative property.Codeine is widely used in cough syrups.

Q177 **Text Solution:**

(4)

Video Solution:



Q178 **Text Solution:**

Infertility refers to the inability to conceive despite regular unprotected intercourse. This condition can arise due to problems in either the male (low sperm count, poor motility, etc.) or the female (ovulation defects, tubal blockage, etc.), so both sexes can be infertile. Hence, the assertion is true. Assisted Reproductive Technologies (ARTs) such as IVF help infertile couples to have children, making the reason statement also true. However, ARTs describe a solution to infertility and do not explain why infertility occurs in both males and females. Therefore, the reason is not the correct explanation of the assertion.

Video Solution:



Q179 **Text Solution:**

The combination of symptoms described—aggressiveness, mood swings, depression, decreased sperm production (due to testicular atrophy), and breast enlargement (gynecomastia) in males—are characteristic side effects of the misuse of **anabolic steroids**.

[New NCERT Class 12th Page No. 144]

Video Solution:



Q180 **Text Solution:**

The assertion is true, but the reason is false because neither of the DNA strands is destroyed during denaturation.

Video Solution:

